

Operator-Theoretic Generalization Bounds for Multitask Deep Learning

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Abstract

We develop operator-theoretic generalization bounds for deep multi-output function classes by representing network layers as Koopman composition operators on vector-valued reproducing kernel Hilbert spaces. In vector-valued Sobolev RKHSs, we derive Rademacher complexity bounds for invertible and width-expanding injective architectures. The estimates separate the output-coupling contribution, represented by the trace of the task matrix, from the layerwise operator norms, Sobolev symbol ratios, determinant factors, and restriction constants generated by the linear maps. We then analyze a distinct one-dimensional Brownian/Cameron–Martin regime. Using the exact anchored derivative-norm characterization of the vector-valued Brownian RKHS, we obtain layerwise bounds for domain-preserving scalar linear maps and anchored diffeomorphic activations; the corresponding factors scale as $|W_l|^{1/2}$ and $\|\sigma'_l\|_\infty^{1/2}$, respectively, and do not involve Sobolev smoothness exponents. Because the Sobolev and Brownian results concern different hypothesis spaces, neither is asserted to dominate the other uniformly. We additionally formulate shared operator learning across tasks, prove a finite-rank representer theorem, derive the exact finite-dimensional problem for squared loss, and establish a target-transfer bound when the learned operator is obtained independently of the target sample. Synthetic and MNIST studies examine stabilized Sobolev-inspired and Brownian-inspired complexity proxies; these empirical proxies are not evaluations of the proved bounds for rank-deficient architectures.

1 Introduction

Understanding the generalization behavior of deep networks remains a central problem in learning theory. Classical approaches control capacity through parameter counts, margins, compression, or norms of the weight matrices (Bartlett and Mendelson, 2002; Mohri et al., 2018; Neyshabur et al., 2015; Golowich et al., 2020; Bartlett et al., 2017; Wei and Ma, 2019; Liu et al., 2024; Arora et al., 2018). Norm-based estimates can avoid an explicit width dependence, but their layerwise factors often retain only coarse information about the geometry of the transformations.

A complementary approach represents each layer by the Koopman operator that it induces on a function space. The closest prior generalization analysis in this direction develops scalar-valued bounds in Sobolev RKHSs and exposes determinant-based volume distortion for full-rank layers (Hashimoto et al., 2024). Two questions then arise naturally. First, how does this construction change for multi-output classes in vector-valued RKHSs? Second, what changes when the underlying function space is not Sobolev but an anchored Brownian/Cameron–Martin RKHS?

Function-space viewpoint. The paper studies both questions at the level of the induced composition operators. In the Sobolev regime, the output coordinates are coupled by a positive-definite task matrix $\mathbf{M}$, and the corresponding RKHS norm is the $\mathbf{M}^{-1}$ -weighted vector Sobolev norm. This makes it possible to separate the task-coupling factor $\text{Tr}(\mathbf{M})$ from the layerwise geometric terms. We treat square invertible maps and non-square injective maps; in the latter case, the proof also records the Sobolev restriction cost created by the lower-dimensional range of each layer.

Brownian/Cameron–Martin viewpoint. For the one-dimensional Brownian kernel, the RKHS norm is exactly an anchored first-derivative energy. This permits a direct change-of-variables analysis without Fourier extensions. Under the domain-preserving conditions needed to keep every composition inside the RKHS, a scalar linear layer contributes $|W|^{1/2}$ and an anchored C^1 diffeomorphism contributes $\|\sigma'\|_\infty^{1/2}$. The Brownian and Sobolev bounds therefore reflect different function-space geometries. They are bounds for different hypothesis classes and are not presented as uniformly comparable numerical estimates.

Contributions. The principal contributions are as follows.

- **Vector-valued Sobolev Koopman bounds.** We extend the Koopman factorization to separable vector-valued Sobolev RKHSs and derive multi-output Rademacher bounds. The output matrix enters through $\text{Tr}(\mathbf{M})$ and the $\mathbf{M}^{-1}$ -weighted terminal norm, while the linear layers contribute operator-norm, Sobolev-symbol, and determinant factors.
- **Injective width expansion.** For injective non-square layers, we state explicit Sobolev trace conditions, define the inverse transpose on the range, and include uniform restriction-operator constants. The rank-deficient case is retained only as a regularized empirical proxy; no bounded Koopman theorem is claimed there.
- **One-dimensional Brownian/Cameron–Martin bound.** We give an exact vector-valued Cameron–Martin characterization and prove the linear and activation composition estimates directly from the derivative norm. This yields a complexity bound without Sobolev smoothness exponents.
- **Shared operator learning and transfer.** We prove a finite-rank representer theorem for a shared Hilbert–Schmidt operator, derive the exact coefficient problem for squared loss, and establish a conditional target bound under source–target sample independence.
- **Empirical proxy study.** Synthetic and MNIST experiments compare stabilized Sobolev-inspired and Brownian-inspired factors. The text distinguishes these proxies from the proved bounds, especially for rectangular or rank-deficient layers.

2 Related Work

Norm-based generalization bounds. A large literature controls deep-network capacity through spectral, Frobenius, path, or reference-matrix norms (Neyshabur et al., 2015; Golowich et al., 2020; Bartlett et al., 2017; Wei and Ma, 2020; Ju et al., 2022). These analyses can avoid direct parameter counting, but their bounds generally summarize each layer through a limited collection of matrix norms. Compression methods can exploit approximate low-rank structure (Arora et al., 2018), while orthogonality-based analyses emphasize conditioning and dynamical stability (Li et al., 2021). The operator-theoretic approach considered here is different: it controls the action of each layer on a specified function space and thereby exposes the volume and regularity distortion of the induced composition operator.

Koopman formulations. Koopman operators encode nonlinear dynamics through linear composition operators on observables. A recent learning-theoretic formulation applies this perspective to scalar deep networks in Sobolev RKHSs and derives determinant-sensitive generalization bounds for full-rank weights (Hashimoto et al., 2024). Our Sobolev results extend that construction to multi-output vector-valued RKHSs and to injective width-expanding maps. Our Brownian result instead uses an anchored one-dimensional Cameron–Martin geometry and therefore requires a separate composition analysis.

Vector-valued RKHSs and multitask learning. Matrix-valued kernels and vector-valued RKHSs provide a standard framework for coupled outputs and multitask learning (Evgeniou et al., 2005; Micchelli and Pontil, 2005; Caponnetto et al., 2008; Argyriou et al., 2007; 2008). Existing guarantees include Rademacher bounds for linear multitask classes (Maurer, 2006), excess-risk bounds for trace-norm regularization (Pontil and Maurer, 2013), representation-learning guarantees (Maurer et al., 2016), and local-complexity analyses (Yousefi et al., 2018). These works characterize task sharing but do not analyze deep compositions through layerwise Koopman operators.

Operator learning and transfer. Representer theorems for learning Koopman operators show how Hilbert-space operator problems can reduce to finite rank (Khosravi, 2023). We use the same general projection principle in a vector-valued multitask setting. The transferable object is a bounded operator between Hilbert function spaces, rather than only an output kernel or a finite-dimensional latent vector.

3 Preliminaries

This section fixes the notation, the vector-valued RKHS conventions, the Rademacher complexity, and the Koopman representation used in the main results.

3.1 Notation and function spaces

The natural numbers are $\mathbb{N} := \{0, 1, 2, \dots\}$ and the positive integers are $\mathbb{N}_{>0} := \{1, 2, \dots\}$. For $n \in \mathbb{N}_{>0}$, write $[n] := \{1, \dots, n\}$. The nonnegative reals are $\mathbb{R}_{\geq 0}$. For a bounded linear operator A between Hilbert spaces, $\text{ran}(A)$, $\ker(A)$, and $\|A\|$ denote its range, kernel, and operator norm. If $\mathbf{W} \in \mathbb{R}^{d \times d'}$ is injective with $d \geq d'$, define its range-volume determinant by

$$|\det(\mathbf{W})| := \det(\mathbf{W}^\top \mathbf{W})^{1/2}.$$

The cone of symmetric positive semidefinite $m \times m$ matrices is $\mathbb{S}_+^m$; $\mathbf{M} \succ 0$ means that $\mathbf{M}$ is strictly positive definite.

A scalar kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is symmetric and positive semidefinite. Its RKHS is denoted by $\mathcal{H}_k$, and $k_{\mathbf{x}} := k(\cdot, \mathbf{x})$ satisfies

$$h(\mathbf{x}) = \langle h, k_{\mathbf{x}} \rangle_{\mathcal{H}_k}.$$

A matrix-valued kernel $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}^{m \times m}$ satisfies $K(\mathbf{x}, \mathbf{x}') = K(\mathbf{x}', \mathbf{x})^\top$ and

$$\sum_{i,j=1}^n \mathbf{y}_i^\top K(\mathbf{x}_i, \mathbf{x}_j) \mathbf{y}_j \geq 0$$

for all $n \in \mathbb{N}_{>0}$, $\mathbf{x}_i \in \mathcal{X}$, and $\mathbf{y}_i \in \mathbb{R}^m$. The induced vvRKHS $\mathcal{H}_K$ satisfies the reproducing identity

$$\langle f, K(\cdot, \mathbf{x}) \mathbf{y} \rangle_{\mathcal{H}_K} = f(\mathbf{x})^\top \mathbf{y}.$$

For a separable kernel $K(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}') \mathbf{M}$ with $\mathbf{M} \succ 0$, the vvRKHS consists of $f = (f_1, \dots, f_m)$ with $f_j \in \mathcal{H}_k$, and

$$\begin{aligned} \langle f, g \rangle_{\mathcal{H}_K} &= \sum_{i,j=1}^m (\mathbf{M}^{-1})_{ij} \langle f_i, g_j \rangle_{\mathcal{H}_k}, \\ \|f\|_{\mathcal{H}_K}^2 &= \sum_{i,j=1}^m (\mathbf{M}^{-1})_{ij} \langle f_i, f_j \rangle_{\mathcal{H}_k}. \end{aligned}$$

The main Sobolev and Brownian theorems assume $\mathbf{M} \succ 0$, so the weighted norm is nondegenerate and no quotient-space convention is needed. The later target-transfer proposition permits $\mathbf{M} \succeq 0$ because its proof uses only the matrix-valued reproducing property and the diagonal trace bound. For every ambient dimension d and order $s > d/2$, let k_s denote the Bessel-potential Sobolev kernel normalized so that its RKHS is $H^s(\mathbb{R}^d)$ with the exact Fourier norm displayed below. The ambient dimension is understood from context. We write

$$K_s(\mathbf{x}, \mathbf{x}') = k_s(\mathbf{x}, \mathbf{x}') \mathbf{M}$$

for the separable matrix-valued Sobolev kernel and denote its vvRKHS by $H_{\mathbf{M}}^s(\mathbb{R}^d, \mathbb{R}^m)$. With the fixed Fourier-transform normalization used to define k_s , its norm is

$$\|h\|_{H_{\mathbf{M}}^s(\mathbb{R}^d, \mathbb{R}^m)}^2 = \int_{\mathbb{R}^d} \left(1 + \|\boldsymbol{\omega}\|_2^2\right)^s \widehat{h}(\boldsymbol{\omega})^* \mathbf{M}^{-1} \widehat{h}(\boldsymbol{\omega}) \, d\boldsymbol{\omega}.$$

For a linear subspace $\mathcal{S} \subset \mathbb{R}^d$, $H_{\mathbf{M}}^s(\mathcal{S}, \mathbb{R}^m)$ is defined after identifying $\mathcal{S}$ isometrically with $\mathbb{R}^{\dim(\mathcal{S})}$. A multi-index is $\alpha \in \mathbb{N}^d$, with $|\alpha| := \sum_{j=1}^d \alpha_j$, and D^α denotes the corresponding weak derivative. Let $\mathcal{H}_{K_1}$ and $\mathcal{H}_{K_2}$ be vvRKHSs on $\mathbb{R}^{d_1}$ and $\mathbb{R}^{d_2}$. For a measurable map $\phi : \mathbb{R}^{d_1} \rightarrow \mathbb{R}^{d_2}$, define

$$\mathcal{D}_\phi := \{g \in \mathcal{H}_{K_2} : g \circ \phi \in \mathcal{H}_{K_1}\}.$$

The Koopman operator is $\mathcal{K}_\phi : \mathcal{D}_\phi \rightarrow \mathcal{H}_{K_1}$, $\mathcal{K}_\phi g := g \circ \phi$. For $\mathbf{a} \in \mathbb{R}^d$, the translation $\mathbf{b}(\mathbf{x}) = \mathbf{x} + \mathbf{a}$ induces $\mathcal{K}_{\mathbf{b}} g = g \circ \mathbf{b}$.

The signed one-dimensional Brownian kernel is

$$k^{(\text{B})}(x, x') := \frac{|x| + |x'| - |x - x'|}{2}.$$

On $\mathcal{X} = [-R, R]$, the separable kernel $K(x, x') = k^{(\text{B})}(x, x') \mathbf{M}$ induces the vector-valued Brownian RKHS $H^{(\text{B})}(\mathcal{X}, \mathbb{R}^m)$. This is the usual anchored Cameron–Martin geometry associated with Brownian covariance kernels ([van der Vaart and van Zanten, 2008](#)); the exact two-sided signed-kernel characterization needed here is proved in [Lemma B.2](#). For a scalar differentiable map σ , $\|\sigma'\|_\infty$ denotes the supremum of $|\sigma'|$ over its stated domain.

3.2 Vector-valued Rademacher complexity

Let $\mathcal{D}_n = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subset \mathcal{X}$ be fixed. For each $i \in [n]$, let $\boldsymbol{\sigma}_i = (\sigma_{i1}, \dots, \sigma_{im})$ have independent coordinates uniformly distributed on $\{-1, +1\}$, and assume that the vectors $\boldsymbol{\sigma}_1, \dots, \boldsymbol{\sigma}_n$ are independent.

Definition 1 (Empirical vector-valued Rademacher complexity). For a class $\mathcal{F}$ of functions $f : \mathcal{X} \rightarrow \mathbb{R}^m$, define

$$\hat{\mathfrak{K}}_n^m(\mathcal{F}) := \mathbb{E}_{\text{Rad}^{mn}} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \left\| \sum_{i=1}^n \langle \sigma_i, f(\mathbf{x}_i) \rangle_{\mathbb{R}^m} \right\| \right].$$

The absolute value is convenient for the operator estimates below. All principal classes are symmetric in the terminal map, so it does not change their order of complexity.

3.3 Koopman representation of deep networks

Fix positive integers $d_0, \dots, d_L$ and m , and let $\mathcal{X} \subset \mathbb{R}^{d_0}$ be the input domain. We consider functions of the form

$$f = g \circ \mathbf{b}_L \circ \mathbf{W}_L \circ \sigma_{L-1} \circ \dots \circ \sigma_1 \circ \mathbf{b}_1 \circ \mathbf{W}_1,$$

where $\mathbf{W}_l \in \mathbb{R}^{d_l \times d_{l-1}}$, $\mathbf{b}_l(\mathbf{x}) = \mathbf{x} + \mathbf{a}_l$ with $\mathbf{a}_l \in \mathbb{R}^{d_l}$, $\sigma_l : \mathbb{R}^{d_l} \rightarrow \mathbb{R}^{d_l}$, and $g : \mathbb{R}^{d_L} \rightarrow \mathbb{R}^m$. The composition order is from right to left. At the function-space level,

$$f = \mathcal{K}_{\mathbf{W}_1} \mathcal{K}_{\mathbf{b}_1} \mathcal{K}_{\sigma_1} \dots \mathcal{K}_{\mathbf{W}_{L-1}} \mathcal{K}_{\mathbf{b}_{L-1}} \mathcal{K}_{\sigma_{L-1}} \mathcal{K}_{\mathbf{W}_L} \mathcal{K}_{\mathbf{b}_L} g. \quad (1)$$

For the Sobolev results, fix $\mathbf{M} \succ 0$ and exponents $s_l > d_l/2$. The space at layer l is $H_{\mathbf{M}}^{s_l}(\mathbb{R}^{d_l}, \mathbb{R}^m)$, induced by $K_{s_l} = k_{s_l} \mathbf{M}$. We make the following standing assumption.

Assumption 1 (Sobolev admissibility). For $l \in [L-1]$, the activation Koopman operator

$$\mathcal{K}_{\sigma_l} : H_{\mathbf{M}}^{s_l}(\mathbb{R}^{d_l}, \mathbb{R}^m) \longrightarrow H_{\mathbf{M}}^{s_l}(\mathbb{R}^{d_l}, \mathbb{R}^m)$$

is bounded. The terminal map belongs to $H_{\mathbf{M}}^{s_L}(\mathbb{R}^{d_L}, \mathbb{R}^m)$. Sufficient smoothness conditions for activation boundedness are given in [Lemma D.1](#).

For $B_g > 0$, let $\mathcal{F}(B_g)$ be the class of all functions represented by the displayed architecture with arbitrary admissible biases, weights satisfying the layer-specific conditions stated later, and terminal maps satisfying

$$\|g\|_{H_{\mathbf{M}}^{s_L}(\mathbb{R}^{d_L}, \mathbb{R}^m)} \leq B_g.$$

We suppress the argument B_g when it is fixed. Smooth compactly supported vector-valued functions provide nontrivial admissible terminal maps.

Assumption 2 (Bounded input kernel diagonal). There exists $\kappa > 0$ such that

$$k_{s_0}(\mathbf{x}, \mathbf{x}) \leq \kappa, \quad \mathbf{x} \in \mathcal{X}.$$

The Koopman factorization isolates the contribution of each layer: translations, activations, and linear maps are bounded separately, and their norms are combined by submultiplicativity. The trace of $\mathbf{M}$ enters through the vector-valued RKHS evaluation term, whereas determinant and Sobolev-symbol factors enter through the linear composition operators.

4 Vector-Valued Sobolev Koopman Bounds

The Sobolev estimates combine a vector-valued RKHS evaluation term with layerwise Koopman operator norms. The task matrix appears through $\text{Tr}(\mathbf{M})$ and the $\mathbf{M}^{-1}$ -weighted terminal norm. The linear maps contribute Sobolev-symbol ratios and determinant terms, while the nonlinearities contribute their composition-operator norms.

4.1 Invertible deep networks

Assume $d_l = d$ for $l = 0, \dots, L$, where $d \in \mathbb{N}_{>0}$, and impose the monotone Sobolev-order condition

$$0 \leq s_0 \leq s_1 \leq \dots \leq s_L, \quad s_l > \frac{d}{2}, \quad l = 0, \dots, L.$$

For an invertible linear map, this monotonicity ensures that every Sobolev-symbol supremum appearing below is finite. For $C, D > 0$, define

$$\mathbf{W}(C, D) := \left\{ \mathbf{W} \in \mathbb{R}^{d \times d} \mid \|\mathbf{W}\| \leq C, \quad |\det(\mathbf{W})| \geq D \right\}.$$

The corresponding class is

$$\mathcal{F}_{\text{inv}} := \left\{ f \in \mathcal{F}(B_g) \mid (\mathbf{W}_1, \dots, \mathbf{W}_L) \in \mathbf{W}(C, D)^L \right\}.$$

Theorem 1 (Invertible Sobolev bound). *Under [Assumptions 1](#) and [2](#) and the monotone Sobolev-order condition above,*

$$\begin{aligned} \hat{\mathfrak{R}}_n^m(\mathcal{F}_{\text{inv}}) &\leq B_g \sqrt{\frac{\kappa \text{Tr}(\mathbf{M})}{n}} \prod_{l=1}^{L-1} \|\mathcal{K}_{\sigma_l}\| \\ &\cdot \sup_{(\mathbf{W}_1, \dots, \mathbf{W}_L) \in \mathbf{W}(C, D)^L} \prod_{l=1}^L \left[\sup_{\boldsymbol{\omega} \in \mathbb{R}^d} \left(\frac{(1 + \|\mathbf{W}_l^\top \boldsymbol{\omega}\|_2^2)^{s_{l-1}}}{(1 + \|\boldsymbol{\omega}\|_2^2)^{s_l}} \right)^{1/2} \frac{1}{|\det(\mathbf{W}_l)|^{1/2}} \right]. \end{aligned}$$

Every translation Koopman operator is an isometry on the weighted Sobolev spaces, so no bias-dependent factor appears.

Corollary 1 (Common Sobolev order). *Suppose $s_l = s > d/2$ for $l = 0, \dots, L$. Then*

$$\hat{\mathfrak{R}}_n^m(\mathcal{F}_{\text{inv}}) \leq B_g \sqrt{\frac{\kappa \text{Tr}(\mathbf{M})}{n}} \left(\frac{\max\{1, C^s\}}{D^{1/2}} \right)^L \prod_{l=1}^{L-1} \|\mathcal{K}_{\sigma_l}\|.$$

4.2 Injective width-expanding networks

Assume $d_0 \leq d_1 \leq \dots \leq d_L$ and set $q_l := d_l - d_{l-1}$. In addition to $s_j > d_j/2$, assume

$$s_l > \frac{q_l}{2}, \quad 0 \leq s_{l-1} \leq s_l - \frac{q_l}{2}, \quad l \in [L].$$

These conditions are sufficient for the standard Sobolev trace map from $\mathbb{R}^{d_l}$ to a d_{l-1} -dimensional linear subspace to be bounded at the stated orders ([Wendland, 2004](#)).

For $C, D > 0$, define

$$\mathbf{W}_l(C, D) := \left\{ \mathbf{W} \in \mathbb{R}^{d_l \times d_{l-1}} \mid \mathbf{W} \text{ is injective, } \|\mathbf{W}\| \leq C, \quad \det(\mathbf{W}^\top \mathbf{W})^{1/2} \geq D \right\}.$$

For $\mathbf{W} \in \mathcal{W}_l(C, D)$, let

$$R_{\mathbf{W}} : H_{\mathbf{M}}^{s_l}(\mathbb{R}^{d_l}, \mathbb{R}^m) \longrightarrow H_{\mathbf{M}}^{s_l-1}(\text{ran}(\mathbf{W}), \mathbb{R}^m), \quad R_{\mathbf{W}}h := h|_{\text{ran}(\mathbf{W})}.$$

Define the uniform restriction constant

$$G_l := \sup_{\mathbf{W} \in \mathcal{W}_l(C, D)} \|R_{\mathbf{W}}\|.$$

The trace theorem and rotational invariance of the Sobolev norm imply $G_l < \infty$.

The injective class is

$$\mathcal{F}_{\text{inj}} := \left\{ f \in \mathcal{F}(B_g) \mid (\mathbf{W}_1, \dots, \mathbf{W}_L) \in \prod_{l=1}^L \mathcal{W}_l(C, D) \right\}.$$

Theorem 2 (Injective Sobolev bound). *Under [Assumptions 1](#) and [2](#) and the trace conditions above,*

$$\begin{aligned} \hat{\mathfrak{R}}_n^m(\mathcal{F}_{\text{inj}}) &\leq B_g \sqrt{\frac{\kappa \text{Tr}(\mathbf{M})}{n}} \prod_{l=1}^{L-1} \|\mathcal{K}_{\sigma_l}\| \\ &\quad \cdot \sup_{(\mathbf{W}_1, \dots, \mathbf{W}_L) \in \prod_{l=1}^L \mathcal{W}_l(C, D)} \prod_{l=1}^L \left[G_l \sup_{\omega \in \text{ran}(\mathbf{W}_l)} \left(\frac{1 + \|\mathbf{W}_l^\top \omega\|_2^2}{1 + \|\omega\|_2^2} \right)^{s_{l-1}/2} \frac{1}{\det(\mathbf{W}_l^\top \mathbf{W}_l)^{1/4}} \right]. \end{aligned}$$

Fix $l \in [L]$, $\mathbf{W}_l \in \mathcal{W}_l(C, D)$, and $h \in H_{\mathbf{M}}^{s_l}(\mathbb{R}^{d_l}, \mathbb{R}^m)$. Define the realized restriction ratio

$$\gamma_l(h, \mathbf{W}_l) := \begin{cases} 0, & h = 0, \\ \frac{\|R_{\mathbf{W}_l}h\|_{H_{\mathbf{M}}^{s_l-1}(\text{ran}(\mathbf{W}_l), \mathbb{R}^m)}}{\|h\|_{H_{\mathbf{M}}^{s_l}(\mathbb{R}^{d_l}, \mathbb{R}^m)}}, & h \neq 0. \end{cases}$$

The factorization $\mathcal{K}_{\mathbf{W}_l} = P_{\mathbf{W}_l} R_{\mathbf{W}_l}$ proved in [Lemma A.6](#) then gives the function-dependent estimate

$$\|\mathcal{K}_{\mathbf{W}_l}h\|_{H_{\mathbf{M}}^{s_l-1}(\mathbb{R}^{d_{l-1}}, \mathbb{R}^m)} \leq \gamma_l(h, \mathbf{W}_l) \sup_{\omega \in \text{ran}(\mathbf{W}_l)} \left(\frac{1 + \|\mathbf{W}_l^\top \omega\|_2^2}{1 + \|\omega\|_2^2} \right)^{s_{l-1}/2} \frac{\|h\|_{H_{\mathbf{M}}^{s_l}(\mathbb{R}^{d_l}, \mathbb{R}^m)}}{\det(\mathbf{W}_l^\top \mathbf{W}_l)^{1/4}}.$$

Moreover,

$$0 \leq \gamma_l(h, \mathbf{W}_l) \leq \|R_{\mathbf{W}_l}\| \leq G_l.$$

In a fixed network, h is the actual downstream function presented to $\mathcal{K}_{\mathbf{W}_l}$ in the Koopman chain. The ratio therefore records the restriction cost of that particular realization. It is function-dependent and cannot replace the uniform constant G_l in the class-level Rademacher theorem.

Remark 1 (Rank-deficient weight matrices). Injectivity is used to obtain a bounded pullback between the stated Sobolev spaces and a nonzero range-volume determinant. For a rank-deficient matrix, the factor $\det(\mathbf{W}^\top \mathbf{W})^{-1/4}$ is singular, and the theorem above does not apply. The stabilized expression $\det(\mathbf{I} + \mathbf{W}^\top \mathbf{W})^{-1/4}$ can be used as a finite empirical surrogate, as in the experiments, but this substitution alone does not prove boundedness of the rank-deficient Koopman operator. A rigorous non-injective extension requires a different function-space or quotient-space analysis and is outside the scope of this paper.

5 Vector-Valued Brownian/Cameron–Martin Koopman Bound

The Brownian result concerns a different hypothesis space from the Sobolev results. We work on the interval $\mathcal{X} = [-R, R]$ with $R > 0$, the signed Brownian kernel

$$k^{(\text{B})}(x, x') = \frac{|x| + |x'| - |x - x'|}{2},$$

and the matrix-valued kernel $K(x, x') = k^{(\text{B})}(x, x') \mathbf{M}$, where $\mathbf{M} \succ 0$. By [Lemma B.2](#),

$$H^{(\text{B})}(\mathcal{X}, \mathbb{R}^m) = \left\{ f : \mathcal{X} \rightarrow \mathbb{R}^m \mid f(0) = 0, \text{ } f \text{ is absolutely continuous, } f' \in L^2(\mathcal{X}, \mathbb{R}^m) \right\},$$

with the exact norm

$$\|f\|_{H^{(\text{B})}(\mathcal{X}, \mathbb{R}^m)}^2 = \int_{-R}^R f'(t)^\top \mathbf{M}^{-1} f'(t) \, dt.$$

For constants $0 < D \leq C \leq 1$, define

$$\mathcal{W}(C, D)_{\text{B}} := \{W \in \mathbb{R} : D \leq |W| \leq C\}.$$

The restriction $C \leq 1$ ensures that $x \mapsto Wx$ maps $\mathcal{X}$ into itself. For $l \in [L - 1]$, assume that $\sigma_l : \mathcal{X} \rightarrow \mathcal{X}$ is a C^1 diffeomorphism satisfying $\sigma_l(0) = 0$. The anchoring condition is necessary because every function in the Brownian RKHS vanishes at the origin. Bias translations are therefore not included in this regime.

Define

$$\mathcal{F}_{\text{inv}}^{(\text{B})} := \left\{ \mathcal{K}_{W_1} \mathcal{K}_{\sigma_1} \cdots \mathcal{K}_{\sigma_{L-1}} \mathcal{K}_{W_L} g \mid \|g\|_{H^{(\text{B})}(\mathcal{X}, \mathbb{R}^m)} \leq B_g, \quad (W_1, \dots, W_L) \in \mathcal{W}(C, D)_{\text{B}}^L \right\}.$$

Theorem 3 (Brownian/Cameron–Martin bound). *For every fixed sample $\{x_1, \dots, x_n\} \subset \mathcal{X}$,*

$$\begin{aligned} \hat{\mathfrak{R}}_n^m(\mathcal{F}_{\text{inv}}^{(\text{B})}) &\leq B_g \sqrt{\frac{R \operatorname{Tr}(\mathbf{M})}{n}} \prod_{l=1}^{L-1} \|\sigma'_l\|_\infty^{1/2} \\ &\quad \cdot \sup_{(W_1, \dots, W_L) \in \mathcal{W}(C, D)_{\text{B}}^L} \prod_{l=1}^L |W_l|^{1/2}. \end{aligned}$$

The layerwise estimates behind the theorem are

$$\begin{aligned} \|\mathcal{K}_W\| &\leq |W|^{1/2}, \\ \|\mathcal{K}_\sigma\| &\leq \|\sigma'\|_\infty^{1/2}. \end{aligned}$$

They follow directly from the anchored derivative norm; no compactly supported extension, Fourier seminorm, or inverse-derivative factor is required.

Remark 2 (Scope of the Brownian comparison). The Sobolev and Brownian results act on different RKHSs, use different domain conditions, and cover different architecture classes. The Brownian theorem is one-dimensional, excludes bias translations, and requires every layer map to preserve $[-R, R]$ and the anchor at zero. It should therefore be interpreted as a separate operator-theoretic regime, not as a uniform improvement over the multidimensional Sobolev bounds.

6 Shared Operator Learning and Transfer

We next consider a shared operator learned from several source tasks. The purpose is to record the Hilbert-space representer structure and the complexity of the target class induced by that operator. The section does not claim that operator sharing improves every transfer problem.

6.1 Shared operator-learning problem

Let $\mathcal{H}_{\text{out}}$ be a Hilbert space of terminal maps. Examples include $H_{\mathbf{M}}^{sL}(\mathbb{R}^{d_L}, \mathbb{R}^m)$ and $H^{(\mathbf{B})}(\mathcal{X}, \mathbb{R}^m)$. Let $\mathcal{H}_K$ be a vvRKHS of functions from $\mathcal{X}$ to $\mathbb{R}^m$, induced by a matrix-valued kernel K . We write

$$\mathcal{L}_2(\mathcal{H}_{\text{out}}, \mathcal{H}_K)$$

for the Hilbert space of Hilbert–Schmidt operators from $\mathcal{H}_{\text{out}}$ to $\mathcal{H}_K$. If $(e_j)_{j \in J}$ is any orthonormal basis of $\mathcal{H}_{\text{out}}$, its inner product and norm are

$$\begin{aligned} \langle S, T \rangle_{\text{HS}} &:= \sum_{j \in J} \langle S e_j, T e_j \rangle_{\mathcal{H}_K}, \\ \|T\|_{\text{HS}}^2 &:= \sum_{j \in J} \|T e_j\|_{\mathcal{H}_K}^2, \end{aligned}$$

and these definitions are independent of the chosen orthonormal basis. We learn an operator

$$T \in \mathcal{L}_2(\mathcal{H}_{\text{out}}, \mathcal{H}_K).$$

For $u \in \mathcal{H}_K$ and $v \in \mathcal{H}_{\text{out}}$, the rank-one operator $u \otimes v$ is defined by

$$(u \otimes v)h = \langle v, h \rangle_{\mathcal{H}_{\text{out}}} u, \quad h \in \mathcal{H}_{\text{out}}.$$

Let $N_s \in \mathbb{N}_{>0}$ be the number of source tasks. For task $t \in [N_s]$, let $g_t \in \mathcal{H}_{\text{out}}$ be its terminal representation and let $(\mathbf{x}_{ti}, \mathbf{y}_{ti}) \in \mathcal{X} \times \mathbb{R}^m$, $i \in [n_t]$, be its training data. For $\lambda > 0$, consider

$$\underset{T \in \mathcal{L}_2(\mathcal{H}_{\text{out}}, \mathcal{H}_K)}{\text{minimize}} \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \ell(\mathbf{y}_{ti}, (T g_t)(\mathbf{x}_{ti})) + \lambda \|T\|_{\text{HS}}^2, \quad (2)$$

where $\ell : \mathbb{R}^m \times \mathbb{R}^m \rightarrow [0, \infty)$ is finite-valued, continuous, and convex in its second argument.

Proposition 1 (Well-posedness of shared operator learning). *Problem (2) has a unique minimizer in $\mathcal{L}_2(\mathcal{H}_{\text{out}}, \mathcal{H}_K)$.*

6.2 Representer theorem

Theorem 4 (Shared-operator representer theorem). *The unique minimizer $\hat{T}$ of (2) has a finite-rank representation*

$$\hat{T} = \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \sum_{a=1}^m c_{tia} (K(\cdot, \mathbf{x}_{ti}) \mathbf{e}_a) \otimes g_t,$$

where $(\mathbf{e}_a)_{a=1}^m$ is the standard basis of $\mathbb{R}^m$. Consequently, for every $g \in \mathcal{H}_{\text{out}}$ and $\mathbf{x} \in \mathcal{X}$,

$$(\hat{T}g)(\mathbf{x}) = \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \sum_{a=1}^m c_{tia} \langle g_t, g \rangle_{\mathcal{H}_{\text{out}}} K(\mathbf{x}, \mathbf{x}_{ti}) \mathbf{e}_a.$$

6.3 Finite-dimensional reduction

Theorem 5 (Squared-loss coefficient problem). *Assume $\ell(\mathbf{y}, \mathbf{z}) = \|\mathbf{y} - \mathbf{z}\|_2^2$. The coefficients of any representer expansion of the unique minimizer in [Theorem 4](#) solve*

$$\begin{aligned} \min_c \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \left\| \mathbf{y}_{ti} - \sum_{t'=1}^{N_s} \sum_{j=1}^{n_{t'}} \sum_{a=1}^m c_{t'ja} \langle g_{t'}, g_t \rangle_{\mathcal{H}_{\text{out}}} K(\mathbf{x}_{ti}, \mathbf{x}_{t'j}) \mathbf{e}_a \right\|_2^2 \\ + \lambda \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \sum_{a=1}^m \sum_{t'=1}^{N_s} \sum_{j=1}^{n_{t'}} \sum_{b=1}^m c_{tia} c_{t'jb} \langle g_t, g_{t'} \rangle_{\mathcal{H}_{\text{out}}} \mathbf{e}_a^\top K(\mathbf{x}_{ti}, \mathbf{x}_{t'j}) \mathbf{e}_b. \end{aligned}$$

Conversely, every coefficient tensor c defines a finite-rank operator of the representer form, so the displayed problem has the same optimal value as [\(2\)](#) under squared loss.

6.4 Target-transfer bound

Let $K(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}') \mathbf{M}$ with $\mathbf{M} \succeq 0$ and $k(\mathbf{x}, \mathbf{x}) \leq \kappa$. Assume that $\hat{T}$ is measurable with respect to the source data and is independent of the target sample. Conditionally on $\hat{T}$, define

$$\mathcal{F}_{\hat{T}}(B) := \left\{ \hat{T}g : \|g\|_{\mathcal{H}_{\text{out}}} \leq B \right\}.$$

Let

$$R_0(f) := \mathbb{E}[\ell(Y, f(X))], \quad R_0^* := \inf_f R_0(f),$$

where the infimum is over all measurable predictors for which the risk is defined. Let $\hat{f}_0$ be a measurable empirical risk minimizer over $\mathcal{F}_{\hat{T}}(B)$ on an independent target sample of size n_0 . Define the approximation error

$$A_{\hat{T}}(B) := \inf_{\|g\|_{\mathcal{H}_{\text{out}}} \leq B} R_0(\hat{T}g) - R_0^*.$$

Proposition 2 (Conditional target bound). *Assume $0 \leq \ell \leq C_\ell$ and that $\ell(y, \cdot)$ is L_ℓ -Lipschitz with respect to the Euclidean norm. Assume also that all suprema appearing in the Rademacher complexities and uniform-deviation quantities used below are measurable. Conditionally on $\hat{T}$, with probability at least $1 - \delta$ over the target sample,*

$$\begin{aligned} R_0(\hat{f}_0) - R_0^* &\leq A_{\hat{T}}(B) + 4\sqrt{2} L_\ell B \|\hat{T}\| \sqrt{\frac{\kappa \text{Tr}(\mathbf{M})}{n_0}} \\ &\quad + 2C_\ell \sqrt{\frac{\log(2/\delta)}{2n_0}}. \end{aligned}$$

The same statement holds with $\|\hat{T}\|_{\text{HS}}$ in place of $\|\hat{T}\|$.

Remark 3. The proposition separates approximation from estimation. It does not assert that a shared operator always improves target performance. It states that, when a useful source-learned operator has controlled norm, the target class induced by its image has controlled complexity. If $\hat{T}$ is itself a Koopman chain satisfying [Theorems 1 to 3](#), its operator norm can be bounded by the corresponding layerwise estimates.

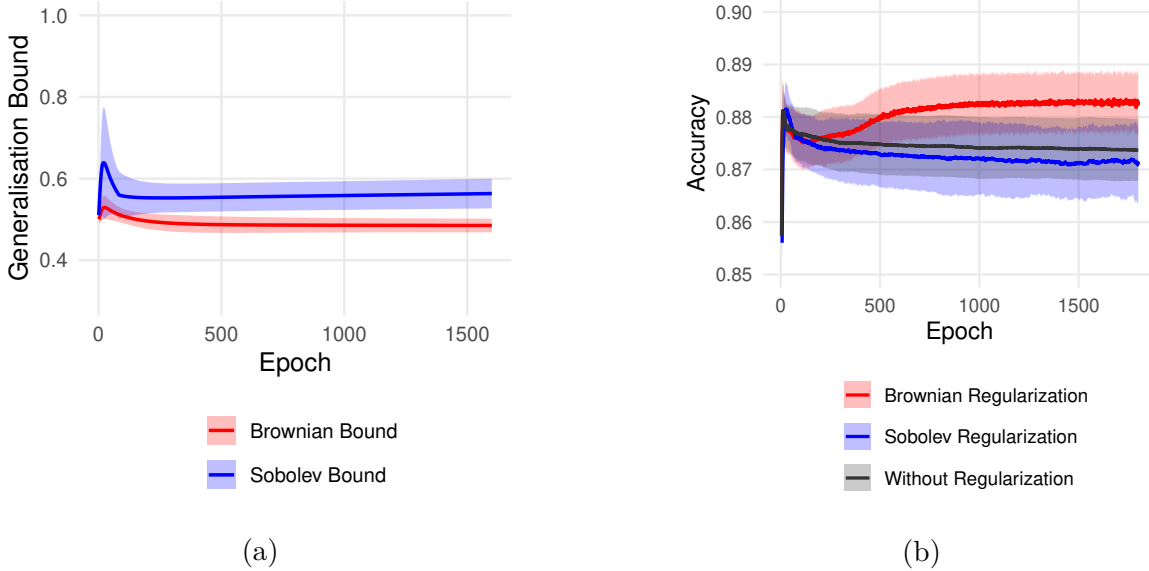


Figure 1: Empirical behavior of stabilized complexity proxies. (a) Sobolev-inspired and Brownian-inspired proxy values during training on the synthetic regression problem. (b) MNIST test performance for the unregularized baseline and for models trained with the two proxy penalties. The labels inside the original plots use the shorthand “bound,” but the quantities are empirical surrogates rather than evaluations of the proved theorems.

7 Empirical Proxy Study

The experiments examine stabilized numerical factors motivated by the layer geometry in the theory. They are not direct evaluations of [Theorems 1 to 3](#): the experimental architectures include rectangular and rank-deficient layers, and the determinant is stabilized by adding the identity.

7.1 Synthetic regression

Following the experimental pattern of [Hashimoto et al. \(2024\)](#), we consider the target

$$t(\mathbf{x}) := \exp\left(-\|\mathbf{2x} - \mathbf{1}\|_2^2\right), \quad \mathbf{x} \in \mathbb{R}^3.$$

The network is

$$f(\mathbf{x}) = g(\mathbf{W}_2 \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2),$$

with $\mathbf{W}_1 \in \mathbb{R}^{3 \times 3}$, $\mathbf{W}_2 \in \mathbb{R}^{6 \times 3}$, $\mathbf{b}_1 \in \mathbb{R}^3$, $\mathbf{b}_2 \in \mathbb{R}^6$, and $g(\mathbf{z}) = \exp(-\|\mathbf{z}\|_2^2)$. The weights use orthogonal initialization ([Saxe et al., 2014](#)), the biases use a uniform initialization, and the activation is the smooth Leaky ReLU of [Biswas et al. \(2022\)](#). Training runs for 1600 epochs with learning rate 3×10^{-3} and an L_2 penalty of 10^{-4} .

For descriptive comparison, define

$$\begin{aligned} \text{SP} &:= \prod_{l=1}^L \frac{\|\mathbf{W}_l\|^{s_l}}{\det(\mathbf{I} + \mathbf{W}_l^\top \mathbf{W}_l)^{1/4}}, & s_l &:= \frac{d_l + 0.1}{2}, \\ \text{BP} &:= \prod_{l=1}^L \frac{\|\mathbf{W}_l\|}{\det(\mathbf{I} + \mathbf{W}_l^\top \mathbf{W}_l)^{1/4}}. \end{aligned}$$

The identity stabilization keeps both proxies finite for rectangular or rank-deficient matrices. It is precisely why these expressions must not be read as the determinant factors in the proved injective theorem. Figure 1(a) reports the trajectories over five initializations. In this experiment, BP is numerically smaller and less variable than SP. This is a descriptive observation about the chosen surrogates, not evidence that the Brownian theorem uniformly dominates the Sobolev theorem. The MNIST regularization experiment and its architecture are described in [Section E](#).

8 Limitations

The Sobolev theorems require invertible or injective linear maps and bounded activation composition operators on the selected weighted Sobolev spaces. The injective theorem additionally requires the stated trace conditions, and its restriction constants may be large. Rank-deficient layers are not covered by replacing a singular determinant with a stabilized one; the experiments use that replacement only as a proxy.

The Brownian theorem is one-dimensional. It requires $|W_t| \leq 1$, activations that map $[-R, R]$ diffeomorphically onto itself, and the anchor condition $\sigma_t(0) = 0$. Bias translations are excluded because they do not preserve the anchored RKHS. The theorem therefore does not directly cover standard unconstrained deep architectures.

All complexity bounds also depend on a prescribed terminal-map norm. The shared-transfer result is conditional on an operator learned independently of the target sample and separates estimation from approximation; it does not guarantee that the source-learned operator is useful for a particular target task.

9 Conclusion

We developed operator-theoretic Rademacher complexity bounds for multi-output compositions in vector-valued RKHSs. In the Sobolev regime, the analysis separates task coupling, terminal-map size, activation composition norms, determinant-based volume distortion, and the restriction cost of injective width expansion. In the one-dimensional Brownian/Cameron–Martin regime, the exact anchored derivative norm gives direct linear and activation estimates without Sobolev exponents or Fourier extensions. We also established a finite-rank shared-operator representer theorem, its squared-loss coefficient reduction, and a source-conditioned target bound. The two function-space regimes have different assumptions and should be compared structurally rather than treated as uniformly ordered numerical bounds.

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A Proofs of the Sobolev Bounds

This appendix proves the vector-valued Sobolev results. Every calculation is carried out in the $\mathbf{M}^{-1}$ -weighted Sobolev norm used in the theorem statements. The proofs first establish a common vvRKHS Rademacher estimate and then derive the layerwise composition bounds.

A.1 A common vvRKHS Rademacher estimate

Lemma A.1 (Operator-image Rademacher bound). *Let $\mathcal{H}_G$ be a Hilbert space and let $\mathcal{H}_K$ be a vvRKHS of functions from $\mathcal{X}$ to $\mathbb{R}^m$ with matrix-valued kernel K . Let $\mathcal{T}$ be a family of bounded linear operators from $\mathcal{H}_G$ to $\mathcal{H}_K$, and assume that*

$$\sup_{T \in \mathcal{T}} \|T\| \leq A.$$

For $B > 0$, define

$$\mathcal{F}_{\mathcal{T}}(B) := \{Tg : T \in \mathcal{T}, \|g\|_{\mathcal{H}_G} \leq B\}.$$

Then, for every fixed sample $\{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subset \mathcal{X}$,

$$\hat{\mathfrak{R}}_n^m(\mathcal{F}_{\mathcal{T}}(B)) \leq \frac{BA}{n} \mathbb{E}_{\text{Rad}^{mn}} \left[\left\| \sum_{i=1}^n K(\cdot, \mathbf{x}_i) \boldsymbol{\sigma}_i \right\|_{\mathcal{H}_K} \right].$$

If $K(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}') \mathbf{M}$ with $\mathbf{M} \succeq 0$ and $k(\mathbf{x}_i, \mathbf{x}_i) \leq \kappa$, then

$$\hat{\mathfrak{R}}_n^m(\mathcal{F}_{\mathcal{T}}(B)) \leq BA \sqrt{\frac{\kappa \text{Tr}(\mathbf{M})}{n}}.$$

Proof. Fix the sample and a realization of the Rademacher vectors. Define

$$V_{\boldsymbol{\sigma}} := \sum_{i=1}^n K(\cdot, \mathbf{x}_i) \boldsymbol{\sigma}_i \in \mathcal{H}_K.$$

For $T \in \mathcal{T}$ and $g \in \mathcal{H}_G$, the reproducing identity gives

$$\begin{aligned} \sum_{i=1}^n \langle \boldsymbol{\sigma}_i, (Tg)(\mathbf{x}_i) \rangle_{\mathbb{R}^m} &= \sum_{i=1}^n \langle Tg, K(\cdot, \mathbf{x}_i) \boldsymbol{\sigma}_i \rangle_{\mathcal{H}_K} \\ &= \langle Tg, V_{\boldsymbol{\sigma}} \rangle_{\mathcal{H}_K}. \end{aligned}$$

The adjoint relation then yields

$$\langle Tg, V_{\boldsymbol{\sigma}} \rangle_{\mathcal{H}_K} = \langle g, T^* V_{\boldsymbol{\sigma}} \rangle_{\mathcal{H}_G}.$$

Applying the Cauchy–Schwarz inequality in $\mathcal{H}_G$ gives

$$\begin{aligned} \left| \langle g, T^* V_{\boldsymbol{\sigma}} \rangle_{\mathcal{H}_G} \right| &\leq \|g\|_{\mathcal{H}_G} \|T^* V_{\boldsymbol{\sigma}}\|_{\mathcal{H}_G} \\ &\leq B \|T^*\| \|V_{\boldsymbol{\sigma}}\|_{\mathcal{H}_K} \\ &= B \|T\| \|V_{\boldsymbol{\sigma}}\|_{\mathcal{H}_K} \\ &\leq BA \|V_{\boldsymbol{\sigma}}\|_{\mathcal{H}_K}. \end{aligned}$$

Taking the supremum over T and g , dividing by n , and then taking the Rademacher expectation proves the first assertion.

Assume now that $K = k\mathbf{M}$ and $k(\mathbf{x}_i, \mathbf{x}_i) \leq \kappa$. Jensen's inequality for the concave square-root function gives

$$\mathbb{E}_{\text{Rad}^{mn}} [\|V_{\boldsymbol{\sigma}}\|_{\mathcal{H}_K}] \leq \left(\mathbb{E}_{\text{Rad}^{mn}} [\|V_{\boldsymbol{\sigma}}\|_{\mathcal{H}_K}^2] \right)^{1/2}.$$

Expanding the squared norm by the reproducing property gives

$$\begin{aligned} \|V_{\boldsymbol{\sigma}}\|_{\mathcal{H}_K}^2 &= \sum_{i=1}^n \sum_{j=1}^n \langle K(\cdot, \mathbf{x}_i) \boldsymbol{\sigma}_i, K(\cdot, \mathbf{x}_j) \boldsymbol{\sigma}_j \rangle_{\mathcal{H}_K} \\ &= \sum_{i=1}^n \sum_{j=1}^n \boldsymbol{\sigma}_i^\top K(\mathbf{x}_i, \mathbf{x}_j) \boldsymbol{\sigma}_j. \end{aligned}$$

When $i \neq j$, independence and centering imply

$$\mathbb{E}_{\text{Rad}^{mn}} [\boldsymbol{\sigma}_i^\top K(\mathbf{x}_i, \mathbf{x}_j) \boldsymbol{\sigma}_j] = 0.$$

For a diagonal term, independence of the coordinates and the identity $\mathbb{E}_{\text{Rad}^{mn}} [\boldsymbol{\sigma}_i \boldsymbol{\sigma}_i^\top] = \mathbf{I}_m$ give

$$\begin{aligned} \mathbb{E}_{\text{Rad}^{mn}} [\boldsymbol{\sigma}_i^\top K(\mathbf{x}_i, \mathbf{x}_i) \boldsymbol{\sigma}_i] &= \text{Tr} \left(K(\mathbf{x}_i, \mathbf{x}_i) \mathbb{E}_{\text{Rad}^{mn}} [\boldsymbol{\sigma}_i \boldsymbol{\sigma}_i^\top] \right) \\ &= \text{Tr} (K(\mathbf{x}_i, \mathbf{x}_i)) \\ &= k(\mathbf{x}_i, \mathbf{x}_i) \text{Tr} (\mathbf{M}) \\ &\leq \kappa \text{Tr} (\mathbf{M}). \end{aligned}$$

Consequently,

$$\mathbb{E}_{\text{Rad}^{mn}} [\|V_{\boldsymbol{\sigma}}\|_{\mathcal{H}_K}^2] \leq n\kappa \text{Tr} (\mathbf{M}).$$

Substitution into the first assertion yields

$$\begin{aligned} \widehat{\mathfrak{R}}_n^m (\mathcal{F}_{\mathcal{T}} (B)) &\leq \frac{BA}{n} \sqrt{n\kappa \text{Tr} (\mathbf{M})} \\ &= BA \sqrt{\frac{\kappa \text{Tr} (\mathbf{M})}{n}}. \end{aligned}$$

This proves the second assertion. □

A.2 Translation and invertible linear composition

Lemma A.2 (Weighted Sobolev translations). *Let $s \geq 0$, $d, m \in \mathbb{N}_{>0}$, $\mathbf{M} \succ 0$, and $\mathbf{a} \in \mathbb{R}^d$. For $\mathbf{b}(\mathbf{x}) = \mathbf{x} + \mathbf{a}$, the Koopman operator*

$$\mathcal{K}_{\mathbf{b}} : H_{\mathbf{M}}^s (\mathbb{R}^d, \mathbb{R}^m) \longrightarrow H_{\mathbf{M}}^s (\mathbb{R}^d, \mathbb{R}^m)$$

is an isometry.

Proof. Let $h \in H_{\mathbf{M}}^s (\mathbb{R}^d, \mathbb{R}^m)$. The Fourier transform of a translation satisfies

$$\widehat{h \circ \mathbf{b}}(\boldsymbol{\omega}) = e^{i\mathbf{a}^\top \boldsymbol{\omega}} \widehat{h}(\boldsymbol{\omega}),$$

where the sign of the phase depends on the Fourier-transform convention and is irrelevant for the norm. Since the phase is scalar and has modulus one,

$$\begin{aligned} \widehat{h \circ \mathbf{b}}(\boldsymbol{\omega})^* \mathbf{M}^{-1} \widehat{h \circ \mathbf{b}}(\boldsymbol{\omega}) &= \left| e^{i\mathbf{a}^\top \boldsymbol{\omega}} \right|^2 \widehat{h}(\boldsymbol{\omega})^* \mathbf{M}^{-1} \widehat{h}(\boldsymbol{\omega}) \\ &= \widehat{h}(\boldsymbol{\omega})^* \mathbf{M}^{-1} \widehat{h}(\boldsymbol{\omega}). \end{aligned}$$

Therefore,

$$\begin{aligned} \|\mathcal{K}_{\mathbf{b}} h\|_{H_{\mathbf{M}}^s (\mathbb{R}^d, \mathbb{R}^m)}^2 &= \int_{\mathbb{R}^d} \left(1 + \|\boldsymbol{\omega}\|_2^2\right)^s \widehat{h \circ \mathbf{b}}(\boldsymbol{\omega})^* \mathbf{M}^{-1} \widehat{h \circ \mathbf{b}}(\boldsymbol{\omega}) \, d\boldsymbol{\omega} \\ &= \int_{\mathbb{R}^d} \left(1 + \|\boldsymbol{\omega}\|_2^2\right)^s \widehat{h}(\boldsymbol{\omega})^* \mathbf{M}^{-1} \widehat{h}(\boldsymbol{\omega}) \, d\boldsymbol{\omega} \\ &= \|h\|_{H_{\mathbf{M}}^s (\mathbb{R}^d, \mathbb{R}^m)}^2. \end{aligned}$$

Thus $\|\mathcal{K}_{\mathbf{b}} h\| = \|h\|$ for every h , and the operator is an isometry. □

Lemma A.3 (Invertible weighted Sobolev pullback). *Let $s_{\text{in}}, s_{\text{out}} \geq 0$, let $\mathbf{M} \succ 0$, and let $\mathbf{W} \in \mathbb{R}^{d \times d}$ be invertible. Then*

$$\mathcal{K}_{\mathbf{W}} : H_{\mathbf{M}}^{s_{\text{out}}}(\mathbb{R}^d, \mathbb{R}^m) \longrightarrow H_{\mathbf{M}}^{s_{\text{in}}}(\mathbb{R}^d, \mathbb{R}^m)$$

is bounded whenever the displayed supremum below is finite, and

$$\|\mathcal{K}_{\mathbf{W}}\| \leq \sup_{\omega \in \mathbb{R}^d} \left(\frac{(1 + \|\mathbf{W}^\top \omega\|_2^2)^{s_{\text{in}}}}{(1 + \|\omega\|_2^2)^{s_{\text{out}}}} \right)^{1/2} \frac{1}{|\det(\mathbf{W})|^{1/2}}.$$

Proof. Let $h \in H_{\mathbf{M}}^{s_{\text{out}}}(\mathbb{R}^d, \mathbb{R}^m)$. The Fourier scaling formula for an invertible linear map gives

$$\widehat{h \circ \mathbf{W}}(\omega) = \frac{1}{|\det(\mathbf{W})|} \widehat{h}(\mathbf{W}^{-\top} \omega).$$

Insert this identity into the weighted Sobolev norm:

$$\begin{aligned} \|\mathcal{K}_{\mathbf{W}} h\|_{H_{\mathbf{M}}^{s_{\text{in}}}(\mathbb{R}^d, \mathbb{R}^m)}^2 &= \int_{\mathbb{R}^d} (1 + \|\omega\|_2^2)^{s_{\text{in}}} \widehat{h \circ \mathbf{W}}(\omega)^* \mathbf{M}^{-1} \widehat{h \circ \mathbf{W}}(\omega) \, d\omega \\ &= \frac{1}{|\det(\mathbf{W})|^2} \int_{\mathbb{R}^d} (1 + \|\omega\|_2^2)^{s_{\text{in}}} \widehat{h}(\mathbf{W}^{-\top} \omega)^* \mathbf{M}^{-1} \widehat{h}(\mathbf{W}^{-\top} \omega) \, d\omega. \end{aligned}$$

Use the change of variables

$$\omega = \mathbf{W}^\top \xi.$$

Its Jacobian is $|\det(\mathbf{W})|$, so

$$d\omega = |\det(\mathbf{W})| \, d\xi.$$

The preceding norm therefore equals

$$\begin{aligned} \|\mathcal{K}_{\mathbf{W}} h\|_{H_{\mathbf{M}}^{s_{\text{in}}}}^2 &= \frac{1}{|\det(\mathbf{W})|} \int_{\mathbb{R}^d} (1 + \|\mathbf{W}^\top \xi\|_2^2)^{s_{\text{in}}} \widehat{h}(\xi)^* \mathbf{M}^{-1} \widehat{h}(\xi) \, d\xi \\ &= \frac{1}{|\det(\mathbf{W})|} \int_{\mathbb{R}^d} \frac{(1 + \|\mathbf{W}^\top \xi\|_2^2)^{s_{\text{in}}}}{(1 + \|\xi\|_2^2)^{s_{\text{out}}}} (1 + \|\xi\|_2^2)^{s_{\text{out}}} \widehat{h}(\xi)^* \mathbf{M}^{-1} \widehat{h}(\xi) \, d\xi. \end{aligned}$$

The scalar quotient is nonnegative. Taking its supremum outside the integral yields

$$\|\mathcal{K}_{\mathbf{W}} h\|_{H_{\mathbf{M}}^{s_{\text{in}}}}^2 \leq \frac{1}{|\det(\mathbf{W})|} \sup_{\xi \in \mathbb{R}^d} \frac{(1 + \|\mathbf{W}^\top \xi\|_2^2)^{s_{\text{in}}}}{(1 + \|\xi\|_2^2)^{s_{\text{out}}}} \|h\|_{H_{\mathbf{M}}^{s_{\text{out}}}}^2.$$

Taking square roots proves the claimed operator-norm estimate. □

A.3 Proof of the invertible theorem and its corollary

Proof of Theorem 1. For a fixed admissible network, define the complete Koopman chain

$$T := \mathcal{K}_{\mathbf{W}_1} \mathcal{K}_{\mathbf{b}_1} \mathcal{K}_{\sigma_1} \cdots \mathcal{K}_{\mathbf{W}_{L-1}} \mathcal{K}_{\mathbf{b}_{L-1}} \mathcal{K}_{\sigma_{L-1}} \mathcal{K}_{\mathbf{W}_L} \mathcal{K}_{\mathbf{b}_L}.$$

By Equation (1), every function in the class has the form $f = Tg$ with

$$\|g\|_{H_{\mathbf{M}}^{sL}(\mathbb{R}^d, \mathbb{R}^m)} \leq B_g.$$

Submultiplicativity of the operator norm gives

$$\|T\| \leq \prod_{l=1}^L \|\mathcal{K}_{\mathbf{W}_l}\| \prod_{l=1}^L \|\mathcal{K}_{\mathbf{b}_l}\| \prod_{l=1}^{L-1} \|\mathcal{K}_{\sigma_l}\|.$$

By Lemma A.2, every translation factor is one. By Lemma A.3, the l th linear factor satisfies

$$\|\mathcal{K}_{\mathbf{W}_l}\| \leq \sup_{\boldsymbol{\omega} \in \mathbb{R}^d} \left(\frac{(1 + \|\mathbf{W}_l^\top \boldsymbol{\omega}\|_2^2)^{s_{l-1}}}{(1 + \|\boldsymbol{\omega}\|_2^2)^{s_l}} \right)^{1/2} \frac{1}{|\det(\mathbf{W}_l)|^{1/2}}.$$

Consequently,

$$\begin{aligned} \|T\| &\leq \prod_{l=1}^{L-1} \|\mathcal{K}_{\sigma_l}\| \\ &\quad \cdot \prod_{l=1}^L \left[\sup_{\boldsymbol{\omega} \in \mathbb{R}^d} \left(\frac{(1 + \|\mathbf{W}_l^\top \boldsymbol{\omega}\|_2^2)^{s_{l-1}}}{(1 + \|\boldsymbol{\omega}\|_2^2)^{s_l}} \right)^{1/2} \frac{1}{|\det(\mathbf{W}_l)|^{1/2}} \right]. \end{aligned}$$

Take the supremum of this estimate over all weight tuples in $\mathcal{W}(C, D)^L$. The input space is the vvRKHS induced by $K_{s_0} = k_{s_0} \mathbf{M}$, and Assumption 2 gives

$$k_{s_0}(\mathbf{x}_i, \mathbf{x}_i) \leq \kappa.$$

Apply Lemma A.1 with the terminal Hilbert space $H_{\mathbf{M}}^{sL}(\mathbb{R}^d, \mathbb{R}^m)$, radius B_g , and the family of complete Koopman chains. The resulting estimate is exactly the bound stated in Theorem 1. $\square$

Proof of Corollary 1. Assume $s_l = s$ for every l . For any $\mathbf{W} \in \mathcal{W}(C, D)$ and any $\boldsymbol{\omega} \in \mathbb{R}^d$,

$$\|\mathbf{W}^\top \boldsymbol{\omega}\|_2 \leq \|\mathbf{W}^\top\| \|\boldsymbol{\omega}\|_2 = \|\mathbf{W}\| \|\boldsymbol{\omega}\|_2 \leq C \|\boldsymbol{\omega}\|_2.$$

It follows that

$$\begin{aligned} 1 + \|\mathbf{W}^\top \boldsymbol{\omega}\|_2^2 &\leq 1 + C^2 \|\boldsymbol{\omega}\|_2^2 \\ &\leq \max\{1, C^2\} (1 + \|\boldsymbol{\omega}\|_2^2). \end{aligned}$$

Raise both sides to the power s , divide by $(1 + \|\boldsymbol{\omega}\|_2^2)^s$, and take square roots. This gives

$$\left(\frac{(1 + \|\mathbf{W}^\top \boldsymbol{\omega}\|_2^2)^s}{(1 + \|\boldsymbol{\omega}\|_2^2)^s} \right)^{1/2} \leq \max\{1, C^s\}.$$

The determinant condition in the definition of $\mathcal{W}(C, D)$ gives

$$\frac{1}{|\det(\mathbf{W})|^{1/2}} \leq \frac{1}{D^{1/2}}.$$

Thus each linear layer contributes at most

$$\frac{\max\{1, C^s\}}{D^{1/2}}.$$

Substituting this uniform estimate into Theorem 1 proves the corollary. $\square$

A.4 Injective linear composition and proof of the injective theorem

Lemma A.4 (Uniform weighted Sobolev restriction). *Fix integers $d_{\text{in}} \leq d_{\text{out}}$ and set $q = d_{\text{out}} - d_{\text{in}}$. Let $s_{\text{out}} > q/2$ and let*

$$0 \leq s_{\text{in}} \leq s_{\text{out}} - \frac{q}{2}.$$

For every d_{in} -dimensional linear subspace $\mathcal{S} \subset \mathbb{R}^{d_{\text{out}}}$, restriction extends uniquely to a bounded operator

$$R_{\mathcal{S}} : H_{\mathbf{M}}^{s_{\text{out}}}(\mathbb{R}^{d_{\text{out}}}, \mathbb{R}^m) \longrightarrow H_{\mathbf{M}}^{s_{\text{in}}}(\mathcal{S}, \mathbb{R}^m).$$

Moreover, the operator norm can be bounded by a constant that depends only on d_{in} , d_{out} , s_{in} , and s_{out} , and not on the orientation of $\mathcal{S}$ or on the positive-definite matrix $\mathbf{M}$.

Proof. The scalar Sobolev trace theorem for Bessel-potential spaces states that restriction to a codimension- q linear subspace extends boundedly as

$$H^{s_{\text{out}}}(\mathbb{R}^{d_{\text{out}}}) \longrightarrow H^{s_{\text{out}}-q/2}(\mathcal{S})$$

whenever $s_{\text{out}} > q/2$. The continuous Sobolev embedding along the same Euclidean subspace gives

$$H^{s_{\text{out}}-q/2}(\mathcal{S}) \longrightarrow H^{s_{\text{in}}}(\mathcal{S})$$

when $s_{\text{in}} \leq s_{\text{out}} - q/2$. Composing these two maps gives a scalar restriction bound

$$\|R_{\mathcal{S}}u\|_{H^{s_{\text{in}}}(\mathcal{S})} \leq C_{\text{tr}}\|u\|_{H^{s_{\text{out}}}(\mathbb{R}^{d_{\text{out}}})}.$$

Because Euclidean Sobolev norms are invariant under orthogonal changes of coordinates, C_{tr} may be chosen independently of the orientation of $\mathcal{S}$.

It remains to pass from the scalar estimate to the $\mathbf{M}^{-1}$ -weighted vector estimate. Let

$$\mathbf{A} := \mathbf{M}^{-1/2}.$$

For a vector-valued function h , define $u = \mathbf{A}h$. Since $\mathbf{A}$ acts only on the output coordinate and restriction acts only on the input variable,

$$\mathbf{A}(R_{\mathcal{S}}h) = R_{\mathcal{S}}(\mathbf{A}h).$$

The weighted Sobolev norm can be written as the ordinary product Sobolev norm of $\mathbf{A}h$:

$$\|h\|_{H_{\mathbf{M}}^s(\mathbb{R}^d, \mathbb{R}^m)}^2 = \sum_{a=1}^m \|(\mathbf{A}h)_a\|_{H^s(\mathbb{R}^d)}^2.$$

Apply the scalar restriction estimate to each component of $\mathbf{A}h$ and sum over a :

$$\begin{aligned} \|R_{\mathcal{S}}h\|_{H_{\mathbf{M}}^{s_{\text{in}}}(\mathcal{S}, \mathbb{R}^m)}^2 &= \sum_{a=1}^m \|R_{\mathcal{S}}(\mathbf{A}h)_a\|_{H^{s_{\text{in}}}(\mathcal{S})}^2 \\ &\leq C_{\text{tr}}^2 \sum_{a=1}^m \|(\mathbf{A}h)_a\|_{H^{s_{\text{out}}}(\mathbb{R}^{d_{\text{out}}})}^2 \\ &= C_{\text{tr}}^2 \|h\|_{H_{\mathbf{M}}^{s_{\text{out}}}(\mathbb{R}^{d_{\text{out}}}, \mathbb{R}^m)}^2. \end{aligned}$$

The constant is the same scalar trace constant and is therefore independent of $\mathbf{M}$. □

Lemma A.5 (Injective weighted Sobolev pullback on the range). *Let $d_{\text{in}} \leq d_{\text{out}}$, let $s \geq 0$, let $\mathbf{M} \succ 0$, and let $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ be injective. Set*

$$\mathcal{S} := \text{ran}(\mathbf{W}).$$

Then the pullback

$$P_{\mathbf{W}} : H_{\mathbf{M}}^s(\mathcal{S}, \mathbb{R}^m) \longrightarrow H_{\mathbf{M}}^s(\mathbb{R}^{d_{\text{in}}}, \mathbb{R}^m), \quad P_{\mathbf{W}}u := u \circ \mathbf{W},$$

is bounded and satisfies

$$\|P_{\mathbf{W}}\| \leq \sup_{\boldsymbol{\omega} \in \mathcal{S}} \left(\frac{1 + \|\mathbf{W}^\top \boldsymbol{\omega}\|_2^2}{1 + \|\boldsymbol{\omega}\|_2^2} \right)^{s/2} \frac{1}{\det(\mathbf{W}^\top \mathbf{W})^{1/4}}.$$

Proof. Choose a matrix $\mathbf{Q} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ whose columns form an orthonormal basis of $\mathcal{S}$. Thus

$$\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_{d_{\text{in}}}, \quad \text{ran}(\mathbf{Q}) = \mathcal{S}.$$

Because every column of $\mathbf{W}$ belongs to $\mathcal{S}$, there is a unique square matrix

$$\mathbf{A} := \mathbf{Q}^\top \mathbf{W} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}$$

such that

$$\mathbf{W} = \mathbf{Q}\mathbf{A}.$$

Injectivity of $\mathbf{W}$ implies invertibility of $\mathbf{A}$. Indeed, if $\mathbf{A}\mathbf{x} = 0$, then

$$\mathbf{W}\mathbf{x} = \mathbf{Q}\mathbf{A}\mathbf{x} = 0,$$

and injectivity of $\mathbf{W}$ forces $\mathbf{x} = 0$.

Identify a function $u : \mathcal{S} \rightarrow \mathbb{R}^m$ with the function

$$\tilde{u} : \mathbb{R}^{d_{\text{in}}} \rightarrow \mathbb{R}^m, \quad \tilde{u}(\mathbf{z}) := u(\mathbf{Q}\mathbf{z}).$$

This identification is isometric by the definition of the Sobolev space on $\mathcal{S}$. Moreover,

$$\begin{aligned} (P_{\mathbf{W}}u)(\mathbf{x}) &= u(\mathbf{W}\mathbf{x}) \\ &= u(\mathbf{Q}\mathbf{A}\mathbf{x}) \\ &= \tilde{u}(\mathbf{A}\mathbf{x}). \end{aligned}$$

Therefore $P_{\mathbf{W}}$ is, in orthonormal range coordinates, exactly the invertible square pullback associated with $\mathbf{A}$. Apply [Lemma A.3](#) with equal input and output order s :

$$\|P_{\mathbf{W}}u\|_{H_{\mathbf{M}}^s(\mathbb{R}^{d_{\text{in}}}, \mathbb{R}^m)} \leq \sup_{\boldsymbol{\xi} \in \mathbb{R}^{d_{\text{in}}}} \left(\frac{1 + \|\mathbf{A}^\top \boldsymbol{\xi}\|_2^2}{1 + \|\boldsymbol{\xi}\|_2^2} \right)^{s/2} \frac{1}{|\det(\mathbf{A})|^{1/2}} \|\tilde{u}\|_{H_{\mathbf{M}}^s(\mathbb{R}^{d_{\text{in}}}, \mathbb{R}^m)}.$$

We now rewrite every factor in coordinate-free form. First,

$$\begin{aligned} \mathbf{W}^\top \mathbf{W} &= \mathbf{A}^\top \mathbf{Q}^\top \mathbf{Q} \mathbf{A} \\ &= \mathbf{A}^\top \mathbf{A}. \end{aligned}$$

Taking determinants gives

$$\begin{aligned}\det(\mathbf{W}^\top \mathbf{W}) &= \det(\mathbf{A}^\top \mathbf{A}) \\ &= \det(\mathbf{A})^2,\end{aligned}$$

and hence

$$|\det(\mathbf{A})|^{1/2} = \det(\mathbf{W}^\top \mathbf{W})^{1/4}.$$

Second, for $\boldsymbol{\xi} \in \mathbb{R}^{d_{\text{in}}}$, set

$$\boldsymbol{\omega} := \mathbf{Q}\boldsymbol{\xi} \in \mathcal{S}.$$

Orthonormality gives

$$\|\boldsymbol{\omega}\|_2 = \|\boldsymbol{\xi}\|_2,$$

and

$$\begin{aligned}\mathbf{W}^\top \boldsymbol{\omega} &= \mathbf{A}^\top \mathbf{Q}^\top \mathbf{Q} \boldsymbol{\xi} \\ &= \mathbf{A}^\top \boldsymbol{\xi}.\end{aligned}$$

As $\boldsymbol{\xi}$ ranges over $\mathbb{R}^{d_{\text{in}}}$, $\boldsymbol{\omega} = \mathbf{Q}\boldsymbol{\xi}$ ranges over $\mathcal{S}$. The square-coordinate supremum is therefore equal to

$$\sup_{\boldsymbol{\omega} \in \mathcal{S}} \left(\frac{1 + \|\mathbf{W}^\top \boldsymbol{\omega}\|_2^2}{1 + \|\boldsymbol{\omega}\|_2^2} \right)^{s/2}.$$

Finally, the isometric identification gives

$$\|\tilde{u}\|_{H_{\mathbf{M}}^s(\mathbb{R}^{d_{\text{in}}}, \mathbb{R}^m)} = \|u\|_{H_{\mathbf{M}}^s(\mathcal{S}, \mathbb{R}^m)}.$$

Substituting these three identities proves the stated bound. $\square$

Lemma A.6 (Complete injective linear-layer estimate). *Under the trace assumptions of [Theorem 2](#), let $\mathbf{W}_l \in \mathcal{W}_l(C, D)$. Then*

$$\mathcal{K}_{\mathbf{W}_l} : H_{\mathbf{M}}^{s_l}(\mathbb{R}^{d_l}, \mathbb{R}^m) \longrightarrow H_{\mathbf{M}}^{s_l-1}(\mathbb{R}^{d_{l-1}}, \mathbb{R}^m)$$

is bounded and

$$\|\mathcal{K}_{\mathbf{W}_l}\| \leq G_l \sup_{\boldsymbol{\omega} \in \text{ran}(\mathbf{W}_l)} \left(\frac{1 + \|\mathbf{W}_l^\top \boldsymbol{\omega}\|_2^2}{1 + \|\boldsymbol{\omega}\|_2^2} \right)^{s_l-1/2} \frac{1}{\det(\mathbf{W}_l^\top \mathbf{W}_l)^{1/4}}.$$

Proof. Let

$$\mathcal{S}_l := \text{ran}(\mathbf{W}_l).$$

For $h \in H_{\mathbf{M}}^{s_l}(\mathbb{R}^{d_l}, \mathbb{R}^m)$, composition with $\mathbf{W}_l$ depends only on the restriction of h to $\mathcal{S}_l$. More precisely,

$$h \circ \mathbf{W}_l = (h|_{\mathcal{S}_l}) \circ \mathbf{W}_l.$$

Thus the full Koopman operator factors as

$$\mathcal{K}_{\mathbf{W}_l} = P_{\mathbf{W}_l} R_{\mathbf{W}_l},$$

where $R_{\mathbf{W}_l}$ is the restriction map defined in the main text and $P_{\mathbf{W}_l}$ is the range pullback from [Lemma A.5](#). Submultiplicativity gives

$$\|\mathcal{K}_{\mathbf{W}_l}\| \leq \|P_{\mathbf{W}_l}\| \|R_{\mathbf{W}_l}\|.$$

By definition of G_l ,

$$\|R_{\mathbf{W}_l}\| \leq G_l.$$

Apply [Lemma A.5](#) with $s = s_{l-1}$ to the other factor. Multiplying the two estimates proves the result. $\square$

Proof of Theorem 2. For a fixed injective network, let T denote the complete Koopman chain in [Equation \(1\)](#). As in the proof of [Theorem 1](#), submultiplicativity gives

$$\|T\| \leq \prod_{l=1}^L \|\mathcal{K}_{\mathbf{W}_l}\| \prod_{l=1}^L \|\mathcal{K}_{\mathbf{b}_l}\| \prod_{l=1}^{L-1} \|\mathcal{K}_{\sigma_l}\|.$$

The dimensions of a translation agree on its domain and codomain, so [Lemma A.2](#) applies at every layer and gives

$$\|\mathcal{K}_{\mathbf{b}_l}\| = 1.$$

For the linear layers, [Lemma A.6](#) gives

$$\|\mathcal{K}_{\mathbf{W}_l}\| \leq G_l \sup_{\boldsymbol{\omega} \in \text{ran}(\mathbf{W}_l)} \left(\frac{1 + \|\mathbf{W}_l^\top \boldsymbol{\omega}\|_2^2}{1 + \|\boldsymbol{\omega}\|_2^2} \right)^{s_{l-1}/2} \frac{1}{\det(\mathbf{W}_l^\top \mathbf{W}_l)^{1/4}}.$$

Consequently,

$$\begin{aligned} \|T\| &\leq \prod_{l=1}^{L-1} \|\mathcal{K}_{\sigma_l}\| \\ &\quad \cdot \prod_{l=1}^L \left[G_l \sup_{\boldsymbol{\omega} \in \text{ran}(\mathbf{W}_l)} \left(\frac{1 + \|\mathbf{W}_l^\top \boldsymbol{\omega}\|_2^2}{1 + \|\boldsymbol{\omega}\|_2^2} \right)^{s_{l-1}/2} \frac{1}{\det(\mathbf{W}_l^\top \mathbf{W}_l)^{1/4}} \right]. \end{aligned}$$

Take the supremum over the product class $\prod_{l=1}^L \mathcal{W}_l(C, D)$. Every function in $\mathcal{F}_{\text{inj}}$ has the form Tg with terminal norm at most B_g . The input vRKHS has kernel $K_{s_0} = k_{s_0} \mathbf{M}$, and its scalar diagonal is bounded by κ under [Assumption 2](#). Applying [Lemma A.1](#) proves the theorem. $\square$

B Proof of the Brownian/Cameron–Martin Bound

B.1 Exact vector-valued Brownian RKHS

Lemma B.1 (Signed-interval feature representation). *For $x \in [-R, R]$, define $\psi_x \in L^2([-R, R])$ by*

$$\psi_x(t) := \begin{cases} \mathbf{1}_{(0,x]}(t), & x \geq 0, \\ -\mathbf{1}_{[x,0)}(t), & x < 0. \end{cases}$$

Then, for all $x, x' \in [-R, R]$,

$$\langle \psi_x, \psi_{x'} \rangle_{L^2([-R, R])} = k^{(\text{B})}(x, x').$$

Proof. There are three sign configurations.

Both points are nonnegative. If $x \geq 0$ and $x' \geq 0$, then the supports are the intervals $(0, x]$ and $(0, x']$. Their intersection has length $\min\{x, x'\}$, so

$$\langle \psi_x, \psi_{x'} \rangle = \min\{x, x'\}.$$

For nonnegative x and x' ,

$$\begin{aligned} \frac{|x| + |x'| - |x - x'|}{2} &= \frac{x + x' - |x - x'|}{2} \\ &= \min\{x, x'\}. \end{aligned}$$

Thus the desired identity holds.

Both points are nonpositive. If $x < 0$ and $x' < 0$, both feature functions carry a minus sign, so their product is positive on the overlap of $[x, 0)$ and $[x', 0)$. The overlap has length $\min\{|x|, |x'|\}$, and hence

$$\langle \psi_x, \psi_{x'} \rangle = \min\{|x|, |x'|\}.$$

Because x and x' have the same negative sign,

$$|x - x'| = ||x| - |x'||.$$

Therefore,

$$\frac{|x| + |x'| - |x - x'|}{2} = \min\{|x|, |x'|\}.$$

Again the desired identity holds.

The points have opposite signs. If one point is nonnegative and the other is negative, the two feature functions have disjoint supports up to the single point zero, which has Lebesgue measure zero. Hence their inner product is zero. In the same sign configuration,

$$|x - x'| = |x| + |x'|,$$

so $k^{(\text{B})}(x, x') = 0$. This completes all cases. $\square$

Lemma B.2 (Exact vector-valued Cameron–Martin characterization). *Let $\mathcal{X} = [-R, R]$, let $\mathbf{M} \succ 0$, and let*

$$K(x, x') = k^{(\text{B})}(x, x') \mathbf{M}.$$

The vvRKHS $H^{(\text{B})}(\mathcal{X}, \mathbb{R}^m)$ is exactly

$$\left\{ f : \mathcal{X} \rightarrow \mathbb{R}^m \mid f(0) = 0, \text{ } f \text{ is absolutely continuous, } f' \in L^2(\mathcal{X}, \mathbb{R}^m) \right\},$$

and its inner product and norm are

$$\begin{aligned} \langle f, h \rangle_{H^{(\text{B})}} &= \int_{-R}^R f'(t)^\top \mathbf{M}^{-1} h'(t) \, dt, \\ \|f\|_{H^{(\text{B})}}^2 &= \int_{-R}^R f'(t)^\top \mathbf{M}^{-1} f'(t) \, dt. \end{aligned}$$

Proof. Define the weighted derivative space

$$L_{\mathbf{M}}^2(\mathcal{X}, \mathbb{R}^m) := \left\{ v : \mathcal{X} \rightarrow \mathbb{R}^m \left| \int_{-R}^R v(t)^\top \mathbf{M}^{-1} v(t) \, dt < \infty \right. \right\},$$

with inner product

$$\langle v, w \rangle_{L_{\mathbf{M}}^2} := \int_{-R}^R v(t)^\top \mathbf{M}^{-1} w(t) \, dt.$$

Let $\lambda_{\min}(\mathbf{M}) > 0$ and $\lambda_{\max}(\mathbf{M}) > 0$ denote the smallest and largest eigenvalues of $\mathbf{M}$. For every $z \in \mathbb{R}^m$,

$$\frac{1}{\lambda_{\max}(\mathbf{M})} \|z\|_2^2 \leq z^\top \mathbf{M}^{-1} z \leq \frac{1}{\lambda_{\min}(\mathbf{M})} \|z\|_2^2.$$

The weighted norm is therefore equivalent to the ordinary vector-valued L^2 norm. Since $L^2(\mathcal{X}, \mathbb{R}^m)$ is complete, $L_{\mathbf{M}}^2(\mathcal{X}, \mathbb{R}^m)$ is a Hilbert space.

Let $\mathcal{C}_{\mathbf{M}}$ denote the anchored absolutely continuous space displayed in the lemma. Define

$$\begin{aligned} J : L_{\mathbf{M}}^2(\mathcal{X}, \mathbb{R}^m) &\longrightarrow \mathcal{C}_{\mathbf{M}}, \\ (Jv)(x) &:= \int_0^x v(t) \, dt, \quad x \in [-R, R], \end{aligned}$$

where, for $x < 0$, the oriented integral means

$$\int_0^x v(t) \, dt = - \int_x^0 v(t) \, dt.$$

For every $v \in L_{\mathbf{M}}^2$, the function Jv is absolutely continuous, satisfies $(Jv)(0) = 0$, and has weak derivative

$$(Jv)' = v$$

almost everywhere. Thus $Jv \in \mathcal{C}_{\mathbf{M}}$. Conversely, if $f \in \mathcal{C}_{\mathbf{M}}$, the fundamental theorem for absolutely continuous functions and the anchoring condition give

$$\begin{aligned} f(x) &= f(0) + \int_0^x f'(t) \, dt \\ &= \int_0^x f'(t) \, dt \\ &= (Jf')(x) \end{aligned}$$

for every $x \in \mathcal{X}$. Hence J is surjective. It is injective because $Jv = 0$ implies $v = (Jv)' = 0$ almost everywhere. Finally,

$$\begin{aligned} \|Jv\|_{\mathcal{C}_{\mathbf{M}}}^2 &= \int_{-R}^R (Jv)'(t)^\top \mathbf{M}^{-1} (Jv)'(t) \, dt \\ &= \int_{-R}^R v(t)^\top \mathbf{M}^{-1} v(t) \, dt \\ &= \|v\|_{L_{\mathbf{M}}^2}^2. \end{aligned}$$

Therefore J is an isometric bijection from the Hilbert space $L_{\mathbf{M}}^2(\mathcal{X}, \mathbb{R}^m)$ onto $\mathcal{C}_{\mathbf{M}}$. In particular, $\mathcal{C}_{\mathbf{M}}$ is complete.

Fix $x \in \mathcal{X}$ and $\mathbf{y} \in \mathbb{R}^m$. The kernel section is

$$K(\cdot, x) \mathbf{y} = k^{(\mathbf{B})}(\cdot, x) \mathbf{M} \mathbf{y}.$$

By [Lemma B.1](#), the weak derivative of the scalar function $t \mapsto k^{(\mathbf{B})}(t, x)$ is $\psi_x(t)$ for almost every t . Consequently,

$$\frac{d}{dt} (K(t, x) \mathbf{y}) = \psi_x(t) \mathbf{M} \mathbf{y}$$

for almost every t . This derivative belongs to L^2 , and the kernel section vanishes at $t = 0$. Hence $K(\cdot, x) \mathbf{y} \in \mathcal{C}_{\mathbf{M}}$.

For $f \in \mathcal{C}_{\mathbf{M}}$, compute its inner product with the kernel section:

$$\begin{aligned} \langle f, K(\cdot, x) \mathbf{y} \rangle_{\mathcal{C}_{\mathbf{M}}} &= \int_{-R}^R f'(t)^\top \mathbf{M}^{-1} (\psi_x(t) \mathbf{M} \mathbf{y}) \, dt \\ &= \int_{-R}^R \psi_x(t) f'(t)^\top \mathbf{y} \, dt \\ &= \left(\int_{-R}^R \psi_x(t) f'(t) \, dt \right)^\top \mathbf{y}. \end{aligned}$$

The signed-interval definition of ψ_x and the fundamental theorem for absolutely continuous functions give

$$\int_{-R}^R \psi_x(t) f'(t) \, dt = f(x) - f(0) = f(x).$$

Therefore,

$$\langle f, K(\cdot, x) \mathbf{y} \rangle_{\mathcal{C}_{\mathbf{M}}} = f(x)^\top \mathbf{y}.$$

Thus K is a reproducing kernel for $\mathcal{C}_{\mathbf{M}}$. By uniqueness of the RKHS associated with a positive semidefinite kernel, $\mathcal{C}_{\mathbf{M}} = H^{(\mathbf{B})}(\mathcal{X}, \mathbb{R}^m)$ isometrically. The displayed inner product and norm are therefore exact. $\square$

B.2 Brownian composition operators

Lemma B.3 (Scalar linear maps in the Brownian RKHS). *Let $W \in \mathbb{R}$ satisfy $|W| \leq 1$. Then*

$$\mathcal{K}_W : H^{(\mathbf{B})}([-R, R], \mathbb{R}^m) \longrightarrow H^{(\mathbf{B})}([-R, R], \mathbb{R}^m)$$

is bounded and

$$\|\mathcal{K}_W\| \leq |W|^{1/2}.$$

Proof. Let $f \in H^{(\mathbf{B})}([-R, R], \mathbb{R}^m)$. Since $|W| \leq 1$, the point Wx belongs to $[-R, R]$ whenever $x \in [-R, R]$. Also,

$$(f \circ W)(0) = f(0) = 0.$$

The chain rule for absolutely continuous functions gives

$$(f \circ W)'(x) = W f'(Wx)$$

for almost every x . By [Lemma B.2](#),

$$\begin{aligned} \|\mathcal{K}_W f\|_{H^{(\mathbb{B})}}^2 &= \int_{-R}^R (W f'(Wx))^\top \mathbf{M}^{-1} (W f'(Wx)) \, dx \\ &= |W|^2 \int_{-R}^R f'(Wx)^\top \mathbf{M}^{-1} f'(Wx) \, dx. \end{aligned}$$

If $W = 0$, the last expression is zero and the result is immediate. Assume $W \neq 0$ and set

$$Q(y) := f'(y)^\top \mathbf{M}^{-1} f'(y).$$

Because $\mathbf{M}^{-1} \succ 0$, one has $Q(y) \geq 0$ for almost every y .

First suppose that $W > 0$. With $y = Wx$ and $dx = dy/W$,

$$\begin{aligned} |W|^2 \int_{-R}^R Q(Wx) \, dx &= W^2 \frac{1}{W} \int_{-WR}^{WR} Q(y) \, dy \\ &= |W| \int_{-|W|R}^{|W|R} Q(y) \, dy. \end{aligned}$$

Next suppose that $W < 0$. The same substitution reverses the integration limits, and therefore

$$\begin{aligned} |W|^2 \int_{-R}^R Q(Wx) \, dx &= W^2 \frac{1}{W} \int_{-WR}^{WR} Q(y) \, dy \\ &= W \int_{|W|R}^{-|W|R} Q(y) \, dy \\ &= -W \int_{-|W|R}^{|W|R} Q(y) \, dy \\ &= |W| \int_{-|W|R}^{|W|R} Q(y) \, dy. \end{aligned}$$

Thus, in both cases,

$$\begin{aligned} \|\mathcal{K}_W f\|_{H^{(\mathbb{B})}}^2 &= |W| \int_{-|W|R}^{|W|R} Q(y) \, dy \\ &\leq |W| \int_{-R}^R Q(y) \, dy \\ &= |W| \|f\|_{H^{(\mathbb{B})}}^2, \end{aligned}$$

where the inequality uses $|W| \leq 1$ and $Q \geq 0$. Taking square roots proves the operator-norm bound. $\square$

Lemma B.4 (Anchored diffeomorphisms in the Brownian RKHS). *Let $\sigma : [-R, R] \rightarrow [-R, R]$ be a C^1 diffeomorphism satisfying $\sigma(0) = 0$. Then*

$$\mathcal{K}_\sigma : H^{(\mathbb{B})}([-R, R], \mathbb{R}^m) \longrightarrow H^{(\mathbb{B})}([-R, R], \mathbb{R}^m)$$

is bounded and

$$\|\mathcal{K}_\sigma\| \leq \|\sigma'\|_\infty^{1/2}.$$

Proof. Let $f \in H^{(\text{B})}([-R, R], \mathbb{R}^m)$. The domain and anchor conditions give

$$(f \circ \sigma)(0) = f(\sigma(0)) = f(0) = 0.$$

The chain rule gives

$$(f \circ \sigma)'(x) = f'(\sigma(x)) \sigma'(x)$$

for almost every x . Therefore,

$$\|\mathcal{K}_\sigma f\|_{H^{(\text{B})}}^2 = \int_{-R}^R |\sigma'(x)|^2 f'(\sigma(x))^\top \mathbf{M}^{-1} f'(\sigma(x)) \, dx.$$

Because σ and σ^{-1} are both C^1 , the chain rule applied to $\sigma^{-1} \circ \sigma = \text{id}$ gives

$$\left(\sigma^{-1}\right)'(\sigma(x)) \sigma'(x) = 1$$

for every $x \in [-R, R]$. Hence $\sigma'(x) \neq 0$ everywhere. Continuity of σ' on the connected interval then implies that σ' has a constant sign, so σ is strictly monotone. Use the change of variables $y = \sigma(x)$. Since σ maps the interval onto itself,

$$dx = \left| \left(\sigma^{-1}\right)'(y) \right| dy = \frac{1}{|\sigma'(\sigma^{-1}(y))|} dy.$$

Substitution yields

$$\begin{aligned} \|\mathcal{K}_\sigma f\|_{H^{(\text{B})}}^2 &= \int_{-R}^R \left| \sigma'(\sigma^{-1}(y)) \right| f'(y)^\top \mathbf{M}^{-1} f'(y) \, dy \\ &\leq \|\sigma'\|_\infty \int_{-R}^R f'(y)^\top \mathbf{M}^{-1} f'(y) \, dy \\ &= \|\sigma'\|_\infty \|f\|_{H^{(\text{B})}}^2. \end{aligned}$$

Taking square roots proves the claim. Notice that the inverse derivative cancels exactly in the change of variables; it is needed only to justify the diffeomorphic change of variables, not as a multiplicative norm factor. $\square$

B.3 Proof of the Brownian theorem

Proof of Theorem 3. For a fixed admissible Brownian network, define

$$T := \mathcal{K}_{W_1} \mathcal{K}_{\sigma_1} \cdots \mathcal{K}_{\sigma_{L-1}} \mathcal{K}_{W_L}.$$

Every function in $\mathcal{F}_{\text{inv}}^{(\text{B})}$ has the form Tg with $\|g\|_{H^{(\text{B})}} \leq B_g$. Submultiplicativity, Lemma B.3, and Lemma B.4 give

$$\begin{aligned} \|T\| &\leq \prod_{l=1}^L \|\mathcal{K}_{W_l}\| \prod_{l=1}^{L-1} \|\mathcal{K}_{\sigma_l}\| \\ &\leq \prod_{l=1}^L |W_l|^{1/2} \prod_{l=1}^{L-1} \|\sigma'_l\|_\infty^{1/2}. \end{aligned}$$

Take the supremum over $(W_1, \dots, W_L) \in \mathcal{W}(C, D)_\text{B}^L$.

The input kernel is $K(x, x') = k^{(\text{B})}(x, x') \mathbf{M}$. On the diagonal,

$$\begin{aligned} k^{(\text{B})}(x, x) &= \frac{|x| + |x| - |x - x|}{2} \\ &= |x| \\ &\leq R. \end{aligned}$$

Apply [Lemma A.1](#) with $\kappa = R$, terminal radius B_g , and the family of complete Brownian Koopman chains. This gives

$$\begin{aligned} \widehat{\mathfrak{K}}_n^m(\mathcal{F}_{\text{inv}}^{(\text{B})}) &\leq B_g \sqrt{\frac{R \text{Tr}(\mathbf{M})}{n}} \prod_{l=1}^{L-1} \|\sigma'_l\|_{\infty}^{1/2} \\ &\quad \cdot \sup_{(W_1, \dots, W_L) \in \mathcal{W}(C, D)_{\text{B}}^L} \prod_{l=1}^L |W_l|^{1/2}, \end{aligned}$$

which is the theorem. □

C Proofs for Shared Operator Learning and Transfer

C.1 Hilbert–Schmidt rank-one identities

Lemma C.1 (Rank-one Hilbert–Schmidt identities). *Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be Hilbert spaces. For $u, u' \in \mathcal{H}_2$ and $v, v' \in \mathcal{H}_1$, define*

$$(u \otimes v)h = \langle v, h \rangle_{\mathcal{H}_1} u.$$

Then $u \otimes v$ is Hilbert–Schmidt and

$$\begin{aligned} \|u \otimes v\|_{\text{HS}} &= \|u\|_{\mathcal{H}_2} \|v\|_{\mathcal{H}_1}, \\ \langle u \otimes v, u' \otimes v' \rangle_{\text{HS}} &= \langle u, u' \rangle_{\mathcal{H}_2} \langle v, v' \rangle_{\mathcal{H}_1}. \end{aligned}$$

Moreover, for every Hilbert–Schmidt operator $T : \mathcal{H}_1 \rightarrow \mathcal{H}_2$,

$$\langle T, u \otimes v \rangle_{\text{HS}} = \langle Tv, u \rangle_{\mathcal{H}_2}.$$

Proof. Let $(e_j)_{j \in J}$ be an orthonormal basis of $\mathcal{H}_1$. By the definition of the Hilbert–Schmidt norm,

$$\begin{aligned} \|u \otimes v\|_{\text{HS}}^2 &= \sum_{j \in J} \|(u \otimes v) e_j\|_{\mathcal{H}_2}^2 \\ &= \sum_{j \in J} \left| \langle v, e_j \rangle_{\mathcal{H}_1} \right|^2 \|u\|_{\mathcal{H}_2}^2 \\ &= \|v\|_{\mathcal{H}_1}^2 \|u\|_{\mathcal{H}_2}^2, \end{aligned}$$

where the last equality is Parseval’s identity. This proves the first formula.

For the inner product of two rank-one operators,

$$\langle u \otimes v, u' \otimes v' \rangle_{\text{HS}} = \sum_{j \in J} \langle (u \otimes v) e_j, (u' \otimes v') e_j \rangle_{\mathcal{H}_2}$$

$$\begin{aligned}
&= \sum_{j \in J} \langle v, e_j \rangle_{\mathcal{H}_1} \overline{\langle v', e_j \rangle_{\mathcal{H}_1}} \langle u, u' \rangle_{\mathcal{H}_2} \\
&= \langle v, v' \rangle_{\mathcal{H}_1} \langle u, u' \rangle_{\mathcal{H}_2}.
\end{aligned}$$

All spaces in the paper are real, so the complex conjugation may be omitted; it is displayed only to emphasize the standard Hilbert-space identity.

Finally,

$$\begin{aligned}
\langle T, u \otimes v \rangle_{\text{HS}} &= \sum_{j \in J} \langle T e_j, (u \otimes v) e_j \rangle_{\mathcal{H}_2} \\
&= \sum_{j \in J} \langle v, e_j \rangle_{\mathcal{H}_1} \langle T e_j, u \rangle_{\mathcal{H}_2} \\
&= \left\langle T \left(\sum_{j \in J} \langle v, e_j \rangle_{\mathcal{H}_1} e_j \right), u \right\rangle_{\mathcal{H}_2} \\
&= \langle T v, u \rangle_{\mathcal{H}_2},
\end{aligned}$$

where convergence follows from the Hilbert–Schmidt property and Parseval’s identity. This proves all three formulas. $\square$

C.2 Well-posedness of the shared operator problem

Proof of Proposition 1. Set

$$\mathcal{E} := \mathcal{L}_2(\mathcal{H}_{\text{out}}, \mathcal{H}_K).$$

This is a Hilbert space under the Hilbert–Schmidt inner product. Define the finite-dimensional prediction space

$$\mathcal{Y}_{\text{tr}} := \prod_{t=1}^{N_s} \prod_{i=1}^{n_t} \mathbb{R}^m$$

with norm

$$\|z\|_{\mathcal{Y}_{\text{tr}}}^2 := \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \|z_{ti}\|_2^2.$$

Define the linear prediction map

$$\begin{aligned}
\Phi : \mathcal{E} &\longrightarrow \mathcal{Y}_{\text{tr}}, \\
(\Phi T)_{ti} &:= (T g_t)(\mathbf{x}_{ti}).
\end{aligned}$$

We first prove that Φ is bounded. Fix $T \in \mathcal{E}$, $t \in [N_s]$, and $i \in [n_t]$. By Euclidean duality and the reproducing property,

$$\begin{aligned}
\|(T g_t)(\mathbf{x}_{ti})\|_2 &= \sup_{\|\mathbf{y}\|_2 \leq 1} \left| \mathbf{y}^\top (T g_t)(\mathbf{x}_{ti}) \right| \\
&= \sup_{\|\mathbf{y}\|_2 \leq 1} \left| \langle T g_t, K(\cdot, \mathbf{x}_{ti}) \mathbf{y} \rangle_{\mathcal{H}_K} \right| \\
&\leq \|T g_t\|_{\mathcal{H}_K} \sup_{\|\mathbf{y}\|_2 \leq 1} \|K(\cdot, \mathbf{x}_{ti}) \mathbf{y}\|_{\mathcal{H}_K}
\end{aligned}$$

$$\leq \|T\| \|g_t\|_{\mathcal{H}_{\text{out}}} \sup_{\|\mathbf{y}\|_2 \leq 1} \|K(\cdot, \mathbf{x}_{ti}) \mathbf{y}\|_{\mathcal{H}_K}.$$

For every $\mathbf{y} \in \mathbb{R}^m$, a second use of the reproducing property gives

$$\begin{aligned} \|K(\cdot, \mathbf{x}_{ti}) \mathbf{y}\|_{\mathcal{H}_K}^2 &= \mathbf{y}^\top K(\mathbf{x}_{ti}, \mathbf{x}_{ti}) \mathbf{y} \\ &\leq \|K(\mathbf{x}_{ti}, \mathbf{x}_{ti})\|_{\text{op}} \|\mathbf{y}\|_2^2. \end{aligned}$$

Since $\|T\| \leq \|T\|_{\text{HS}}$, it follows that

$$\|(Tg_t)(\mathbf{x}_{ti})\|_2^2 \leq \|K(\mathbf{x}_{ti}, \mathbf{x}_{ti})\|_{\text{op}} \|g_t\|_{\mathcal{H}_{\text{out}}}^2 \|T\|_{\text{HS}}^2.$$

Define the finite constant

$$C_\Phi^2 := \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \|K(\mathbf{x}_{ti}, \mathbf{x}_{ti})\|_{\text{op}} \|g_t\|_{\mathcal{H}_{\text{out}}}^2.$$

Summing the preceding pointwise estimate yields

$$\|\Phi T\|_{\mathcal{Y}_{\text{tr}}} \leq C_\Phi \|T\|_{\text{HS}}.$$

Thus Φ is bounded.

Define the finite-dimensional loss functional

$$\mathcal{L}(z) := \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \ell(\mathbf{y}_{ti}, z_{ti}), \quad z \in \mathcal{Y}_{\text{tr}},$$

and define the objective

$$J(T) := \mathcal{L}(\Phi T) + \lambda \|T\|_{\text{HS}}^2.$$

Because ℓ is finite-valued, continuous, convex in its second argument, and nonnegative, $\mathcal{L}$ is finite-valued, continuous, convex, and nonnegative on the finite-dimensional space $\mathcal{Y}_{\text{tr}}$. Consequently,

$$J(T) \geq \lambda \|T\|_{\text{HS}}^2, \quad T \in \mathcal{E}.$$

Let

$$j_\star := \inf_{T \in \mathcal{E}} J(T).$$

The value $j_\star$ is finite because $0 \leq j_\star \leq J(0) < \infty$. Choose a minimizing sequence $(T_r)_{r \in \mathbb{N}_{>0}}$ satisfying

$$J(T_r) \leq j_\star + \frac{1}{r}.$$

The coercive lower bound gives

$$\lambda \|T_r\|_{\text{HS}}^2 \leq J(T_r) \leq j_\star + 1,$$

so the sequence is bounded in the Hilbert space $\mathcal{E}$. Hilbert spaces are reflexive; therefore, after passing to a subsequence without changing notation, there is $T_\star \in \mathcal{E}$ such that

$$T_r \rightharpoonup T_\star$$

weakly in $\mathcal{E}$. Bounded linear maps preserve weak convergence, so

$$\Phi T_r \rightharpoonup \Phi T_\star$$

weakly in $\mathcal{Y}_{\text{tr}}$. Since $\mathcal{Y}_{\text{tr}}$ is finite-dimensional, weak convergence and norm convergence coincide. Hence

$$\Phi T_r \longrightarrow \Phi T_\star$$

in $\mathcal{Y}_{\text{tr}}$. Continuity of $\mathcal{L}$ gives

$$\mathcal{L}(\Phi T_r) \longrightarrow \mathcal{L}(\Phi T_\star).$$

The Hilbert–Schmidt norm is weakly lower semicontinuous, so

$$\|T_\star\|_{\text{HS}}^2 \leq \liminf_{r \rightarrow \infty} \|T_r\|_{\text{HS}}^2.$$

Combining the last two relations gives

$$\begin{aligned} J(T_\star) &= \mathcal{L}(\Phi T_\star) + \lambda \|T_\star\|_{\text{HS}}^2 \\ &\leq \liminf_{r \rightarrow \infty} [\mathcal{L}(\Phi T_r) + \lambda \|T_r\|_{\text{HS}}^2] \\ &= j_\star. \end{aligned}$$

By the definition of $j_\star$, the reverse inequality $j_\star \leq J(T_\star)$ holds. Thus $J(T_\star) = j_\star$, and a minimizer exists.

It remains to prove uniqueness. Let $S, T \in \mathcal{E}$ with $S \neq T$, and let $\theta \in (0, 1)$. Convexity of $\mathcal{L}$ and linearity of Φ give

$$\mathcal{L}(\Phi(\theta S + (1 - \theta)T)) \leq \theta \mathcal{L}(\Phi S) + (1 - \theta) \mathcal{L}(\Phi T).$$

The Hilbert-space norm identity gives

$$\begin{aligned} \|\theta S + (1 - \theta)T\|_{\text{HS}}^2 &= \theta \|S\|_{\text{HS}}^2 + (1 - \theta) \|T\|_{\text{HS}}^2 \\ &\quad - \theta(1 - \theta) \|S - T\|_{\text{HS}}^2. \end{aligned}$$

Because $\lambda > 0$, $\theta(1 - \theta) > 0$, and $S \neq T$, the final term is strictly negative. Therefore,

$$J(\theta S + (1 - \theta)T) < \theta J(S) + (1 - \theta) J(T).$$

Thus J is strictly convex and can have at most one minimizer. Existence and uniqueness together prove the proposition. $\square$

C.3 Proof of the shared-operator representer theorem

Proof of Theorem 4. For $t \in [N_s]$, $i \in [n_t]$, and $a \in [m]$, define the Hilbert–Schmidt operator

$$\Xi_{tia} := (K(\cdot, \mathbf{x}_{ti}) \mathbf{e}_a) \otimes g_t.$$

Let

$$\mathcal{V} := \text{span} \{ \Xi_{tia} : 1 \leq t \leq N_s, 1 \leq i \leq n_t, 1 \leq a \leq m \}$$

inside the Hilbert space $\mathcal{L}_2(\mathcal{H}_{\text{out}}, \mathcal{H}_K)$. The span is finite-dimensional and therefore closed. Fix an arbitrary Hilbert–Schmidt operator T . Decompose it orthogonally as

$$T = T_{\mathcal{V}} + T_{\mathcal{V}^\perp}, \quad T_{\mathcal{V}} \in \mathcal{V}, \quad T_{\mathcal{V}^\perp} \in \mathcal{V}^\perp.$$

We first show that every training prediction depends only on $T_{\mathcal{V}}$. For a fixed triple (t, i, a) , the reproducing property gives

$$\mathbf{e}_a^\top (Tg_t)(\mathbf{x}_{ti}) = \langle Tg_t, K(\cdot, \mathbf{x}_{ti}) \mathbf{e}_a \rangle_{\mathcal{H}_K}.$$

Apply the final identity of [Lemma C.1](#) with

$$u = K(\cdot, \mathbf{x}_{ti}) \mathbf{e}_a, \quad v = g_t.$$

It gives

$$\begin{aligned} \mathbf{e}_a^\top (Tg_t)(\mathbf{x}_{ti}) &= \langle T, \Xi_{tia} \rangle_{\text{HS}} \\ &= \langle T_{\mathcal{V}}, \Xi_{tia} \rangle_{\text{HS}} + \langle T_{\mathcal{V}^\perp}, \Xi_{tia} \rangle_{\text{HS}}. \end{aligned}$$

Since $\Xi_{tia} \in \mathcal{V}$ and $T_{\mathcal{V}^\perp} \perp \mathcal{V}$, the last inner product is zero. Hence

$$\mathbf{e}_a^\top (Tg_t)(\mathbf{x}_{ti}) = \mathbf{e}_a^\top (T_{\mathcal{V}}g_t)(\mathbf{x}_{ti})$$

for every coordinate a . Therefore,

$$(Tg_t)(\mathbf{x}_{ti}) = (T_{\mathcal{V}}g_t)(\mathbf{x}_{ti})$$

for every training pair.

The loss part of the objective in [Equation \(2\)](#) is consequently unchanged when T is replaced by $T_{\mathcal{V}}$. Orthogonality gives the Pythagorean identity

$$\|T\|_{\text{HS}}^2 = \|T_{\mathcal{V}}\|_{\text{HS}}^2 + \|T_{\mathcal{V}^\perp}\|_{\text{HS}}^2.$$

Because $\lambda > 0$, the regularization term is strictly smaller after deleting a nonzero orthogonal component. Hence every minimizer must satisfy

$$T_{\mathcal{V}^\perp} = 0.$$

Thus every minimizer belongs to $\mathcal{V}$. By the definition of $\mathcal{V}$, there are coefficients $c_{tia} \in \mathbb{R}$ such that

$$\hat{T} = \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \sum_{a=1}^m c_{tia} (K(\cdot, \mathbf{x}_{ti}) \mathbf{e}_a) \otimes g_t.$$

The coefficients need not be unique if the spanning operators are linearly dependent, but the represented operator is the minimizer.

To obtain the prediction formula, let $g \in \mathcal{H}_{\text{out}}$. By the definition of a rank-one operator,

$$\hat{T}g = \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \sum_{a=1}^m c_{tia} \langle g_t, g \rangle_{\mathcal{H}_{\text{out}}} K(\cdot, \mathbf{x}_{ti}) \mathbf{e}_a.$$

Evaluating at $\mathbf{x} \in \mathcal{X}$ yields

$$(\hat{T}g)(\mathbf{x}) = \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \sum_{a=1}^m c_{tia} \langle g_t, g \rangle_{\mathcal{H}_{\text{out}}} K(\mathbf{x}, \mathbf{x}_{ti}) \mathbf{e}_a,$$

which proves the theorem. $\square$

C.4 Proof of the finite-dimensional reduction

Proof of Theorem 5. Let a finite-rank operator be represented as

$$T_c = \sum_{t'=1}^{N_s} \sum_{j=1}^{n_{t'}} \sum_{a=1}^m c_{t'ja} (K(\cdot, \mathbf{x}_{t'j}) \mathbf{e}_a) \otimes g_{t'}.$$

For source task t and sample point $\mathbf{x}_{ti}$, apply the definition of the rank-one operators:

$$T_c g_t = \sum_{t'=1}^{N_s} \sum_{j=1}^{n_{t'}} \sum_{a=1}^m c_{t'ja} \langle g_{t'}, g_t \rangle_{\mathcal{H}_{\text{out}}} K(\cdot, \mathbf{x}_{t'j}) \mathbf{e}_a.$$

Evaluation at $\mathbf{x}_{ti}$ gives

$$(T_c g_t)(\mathbf{x}_{ti}) = \sum_{t'=1}^{N_s} \sum_{j=1}^{n_{t'}} \sum_{a=1}^m c_{t'ja} \langle g_{t'}, g_t \rangle_{\mathcal{H}_{\text{out}}} K(\mathbf{x}_{ti}, \mathbf{x}_{t'j}) \mathbf{e}_a.$$

Under squared loss, substituting this expression gives the first term in the coefficient objective stated in the theorem.

It remains to calculate the Hilbert–Schmidt norm. Expand the squared norm using bilinearity:

$$\begin{aligned} \|T_c\|_{\text{HS}}^2 &= \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \sum_{a=1}^m \sum_{t'=1}^{N_s} \sum_{j=1}^{n_{t'}} \sum_{b=1}^m c_{tia} c_{t'jb} \\ &\quad \cdot \langle (K(\cdot, \mathbf{x}_{ti}) \mathbf{e}_a) \otimes g_t, (K(\cdot, \mathbf{x}_{t'j}) \mathbf{e}_b) \otimes g_{t'} \rangle_{\text{HS}}. \end{aligned}$$

Apply the rank-one inner-product identity from Lemma C.1:

$$\begin{aligned} \|T_c\|_{\text{HS}}^2 &= \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \sum_{a=1}^m \sum_{t'=1}^{N_s} \sum_{j=1}^{n_{t'}} \sum_{b=1}^m c_{tia} c_{t'jb} \langle g_t, g_{t'} \rangle_{\mathcal{H}_{\text{out}}} \\ &\quad \cdot \langle K(\cdot, \mathbf{x}_{ti}) \mathbf{e}_a, K(\cdot, \mathbf{x}_{t'j}) \mathbf{e}_b \rangle_{\mathcal{H}_K}. \end{aligned}$$

The vvrKHS reproducing property gives

$$\langle K(\cdot, \mathbf{x}_{ti}) \mathbf{e}_a, K(\cdot, \mathbf{x}_{t'j}) \mathbf{e}_b \rangle_{\mathcal{H}_K} = \mathbf{e}_a^\top K(\mathbf{x}_{ti}, \mathbf{x}_{t'j}) \mathbf{e}_b.$$

Therefore,

$$\|T_c\|_{\text{HS}}^2 = \sum_{t=1}^{N_s} \sum_{i=1}^{n_t} \sum_{a=1}^m \sum_{t'=1}^{N_s} \sum_{j=1}^{n_{t'}} \sum_{b=1}^m c_{tia} c_{t'jb} \langle g_t, g_{t'} \rangle_{\mathcal{H}_{\text{out}}} \mathbf{e}_a^\top K(\mathbf{x}_{ti}, \mathbf{x}_{t'j}) \mathbf{e}_b.$$

Multiplication by λ gives the regularization term in the theorem.

By Theorem 4, every minimizer of the infinite-dimensional problem has at least one coefficient representation of the displayed form. Conversely, every coefficient tensor defines a Hilbert–Schmidt finite-rank operator T_c . The prediction and norm calculations above show that the value of the infinite-dimensional objective at T_c is exactly the displayed coefficient objective. Hence the two optimization problems have the same feasible objective values on the representer subspace, and the representer theorem shows that this subspace contains every minimizer. Their optimal values are therefore equal, and the coefficients of a minimizer solve the stated finite-dimensional problem. $\square$

C.5 Rademacher complexity of a fixed operator image

Lemma C.2 (Conditional target-class complexity). *Let $K(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}') \mathbf{M}$ with $\mathbf{M} \succeq 0$ and $k(\mathbf{x}, \mathbf{x}) \leq \kappa$. Fix a bounded operator $T : \mathcal{H}_{\text{out}} \rightarrow \mathcal{H}_K$ and define*

$$\mathcal{F}_T(B) = \{Tg : \|g\|_{\mathcal{H}_{\text{out}}} \leq B\}.$$

For every sample of size n_0 ,

$$\widehat{\mathfrak{R}}_{n_0}^m(\mathcal{F}_T(B)) \leq B\|T\| \sqrt{\frac{\kappa \text{Tr}(\mathbf{M})}{n_0}}.$$

The same bound holds with $\|T\|_{\text{HS}}$ in place of $\|T\|$.

Proof. Apply [Lemma A.1](#) to the singleton operator family $\mathcal{T} = \{T\}$. This gives the estimate with $\|T\|$. Since every Hilbert–Schmidt operator satisfies

$$\|T\| \leq \|T\|_{\text{HS}},$$

the second estimate follows immediately. $\square$

C.6 A two-sided expected-Rademacher deviation lemma

Lemma C.3 (Two-sided expected-Rademacher deviation). *Let $Z_1, \dots, Z_n$ be independent and identically distributed random variables with values in a measurable space $\mathcal{Z}$, let $S = (Z_1, \dots, Z_n)$, and let $\mathcal{G}$ be a class of measurable functions $g : \mathcal{Z} \rightarrow [0, C]$. Assume that the suprema below are measurable. Define*

$$\begin{aligned} Pg &:= \mathbb{E}[g(Z_1)], \\ P_S g &:= \frac{1}{n} \sum_{i=1}^n g(Z_i), \\ \widehat{\mathfrak{R}}_S(\mathcal{G}) &:= \mathbb{E}_\varepsilon \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(Z_i) \right], \\ \mathfrak{R}_n(\mathcal{G}) &:= \mathbb{E}_S [\widehat{\mathfrak{R}}_S(\mathcal{G})], \end{aligned}$$

where $\varepsilon_1, \dots, \varepsilon_n$ are independent Rademacher variables independent of S . Then, with probability at least $1 - \delta$, the following inequalities hold simultaneously for every $g \in \mathcal{G}$:

$$\begin{aligned} Pg &\leq P_S g + 2\mathfrak{R}_n(\mathcal{G}) + C \sqrt{\frac{\log(2/\delta)}{2n}}, \\ P_S g &\leq Pg + 2\mathfrak{R}_n(\mathcal{G}) + C \sqrt{\frac{\log(2/\delta)}{2n}}. \end{aligned}$$

Proof. Define

$$\begin{aligned} \Phi_+(S) &:= \sup_{g \in \mathcal{G}} (Pg - P_S g), \\ \Phi_-(S) &:= \sup_{g \in \mathcal{G}} (P_S g - Pg). \end{aligned}$$

We first bound the expectations of these two random variables.

Let $S' = (Z'_1, \dots, Z'_n)$ be an independent copy of S . Since $Pg = \mathbb{E}_{S'} [P_{S'}g]$, Jensen's inequality for the supremum gives

$$\begin{aligned} \mathbb{E}_S [\Phi_+(S)] &= \mathbb{E}_S \left[\sup_{g \in \mathcal{G}} (\mathbb{E}_{S'} [P_{S'}g] - P_Sg) \right] \\ &\leq \mathbb{E}_{S, S'} \left[\sup_{g \in \mathcal{G}} (P_{S'}g - P_Sg) \right] \\ &= \mathbb{E}_{S, S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n (g(Z'_i) - g(Z_i)) \right]. \end{aligned}$$

For every sign vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)$, swap Z_i and Z'_i whenever $\varepsilon_i = -1$. Because each pair (Z_i, Z'_i) is exchangeable and the pairs are independent, the joint distribution is unchanged. Therefore,

$$\begin{aligned} \mathbb{E}_{S, S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n (g(Z'_i) - g(Z_i)) \right] \\ = \mathbb{E}_{S, S', \varepsilon} \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (g(Z'_i) - g(Z_i)) \right]. \end{aligned}$$

For every fixed realization of S, S' , and ε ,

$$\begin{aligned} \sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (g(Z'_i) - g(Z_i)) \\ \leq \sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(Z'_i) + \sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n (-\varepsilon_i) g(Z_i). \end{aligned}$$

Taking expectations, using that S' has the same distribution as S , and using that $-\varepsilon$ has the same distribution as ε , we obtain

$$\mathbb{E}_S [\Phi_+(S)] \leq 2\mathfrak{R}_n(\mathcal{G}).$$

The reverse deviation is handled explicitly in the same way. Since $Pg = \mathbb{E}_{S'} [P_{S'}g]$,

$$\begin{aligned} \mathbb{E}_S [\Phi_-(S)] &= \mathbb{E}_S \left[\sup_{g \in \mathcal{G}} (P_Sg - \mathbb{E}_{S'} [P_{S'}g]) \right] \\ &\leq \mathbb{E}_{S, S'} \left[\sup_{g \in \mathcal{G}} (P_Sg - P_{S'}g) \right]. \end{aligned}$$

Interchanging the names of S and S' shows that the final expectation is equal to the ghost-sample expectation already bounded above. Hence

$$\mathbb{E}_S [\Phi_-(S)] \leq 2\mathfrak{R}_n(\mathcal{G}).$$

We next apply bounded differences. If one coordinate Z_i is replaced by an arbitrary value $\tilde{Z}_i$, then, for every $g \in \mathcal{G}$,

$$|P_Sg - P_{\tilde{S}}g| = \frac{1}{n} |g(Z_i) - g(\tilde{Z}_i)|$$

$$\leq \frac{C}{n}.$$

Taking suprema shows that changing one sample coordinate changes each of $\Phi_+(S)$ and $\Phi_-(S)$ by at most C/n . McDiarmid's inequality therefore gives

$$\begin{aligned} \mathbb{P}\left(\Phi_+(S) > \mathbb{E}[\Phi_+(S)] + C\sqrt{\frac{\log(2/\delta)}{2n}}\right) &\leq \frac{\delta}{2}, \\ \mathbb{P}\left(\Phi_-(S) > \mathbb{E}[\Phi_-(S)] + C\sqrt{\frac{\log(2/\delta)}{2n}}\right) &\leq \frac{\delta}{2}. \end{aligned}$$

A union bound, together with the two expectation estimates, implies that with probability at least $1 - \delta$,

$$\begin{aligned} \Phi_+(S) &\leq 2\mathfrak{R}_n(\mathcal{G}) + C\sqrt{\frac{\log(2/\delta)}{2n}}, \\ \Phi_-(S) &\leq 2\mathfrak{R}_n(\mathcal{G}) + C\sqrt{\frac{\log(2/\delta)}{2n}}. \end{aligned}$$

Unpacking the definitions of Φ_+ and Φ_- gives the two simultaneous inequalities in the lemma. $\square$

C.7 Proof of the conditional target-transfer bound

Proof of Proposition 2. Condition on the source data and hence on $\hat{T}$. Under the independence assumption, the target observations remain independent and identically distributed after conditioning, while $\hat{T}$ is fixed. We may therefore carry out the complete argument conditionally for the fixed class

$$\mathcal{F}_{\hat{T}}(B) = \left\{ \hat{T}g : \|g\|_{\mathcal{H}_{\text{out}}} \leq B \right\}.$$

Let

$$S_0 := ((X_1, Y_1), \dots, (X_{n_0}, Y_{n_0}))$$

denote the target sample. Define the empirical target risk by

$$\hat{R}_0(f) := \frac{1}{n_0} \sum_{i=1}^{n_0} \ell(Y_i, f(X_i)),$$

and define the scalar loss class

$$\mathcal{G}_{\hat{T}} := \left\{ (x, y) \mapsto \ell(y, f(x)) : f \in \mathcal{F}_{\hat{T}}(B) \right\}.$$

For a fixed realization of S_0 , its empirical scalar Rademacher complexity is

$$\hat{\mathfrak{R}}_{S_0}(\mathcal{G}_{\hat{T}}) := \mathbb{E}_{\varepsilon} \left[\sup_{f \in \mathcal{F}_{\hat{T}}(B)} \frac{1}{n_0} \sum_{i=1}^{n_0} \varepsilon_i \ell(Y_i, f(X_i)) \right].$$

For each i , set $c_i := \ell(Y_i, 0)$. Because c_i is independent of f , for every fixed Rademacher vector,

$$\sup_{f \in \mathcal{F}_{\hat{T}}(B)} \frac{1}{n_0} \sum_{i=1}^{n_0} \varepsilon_i \ell(Y_i, f(X_i))$$

$$= \frac{1}{n_0} \sum_{i=1}^{n_0} \varepsilon_i c_i + \sup_{f \in \mathcal{F}_{\hat{T}}(B)} \frac{1}{n_0} \sum_{i=1}^{n_0} \varepsilon_i [\ell(Y_i, f(X_i)) - c_i].$$

Taking the Rademacher expectation and using

$$\mathbb{E}_\varepsilon \left[\frac{1}{n_0} \sum_{i=1}^{n_0} \varepsilon_i c_i \right] = 0$$

shows that subtracting $\ell(Y_i, 0)$ leaves the empirical Rademacher complexity unchanged. For each fixed observation (X_i, Y_i) , the map

$$\mathbf{z} \mapsto \ell(Y_i, \mathbf{z}) - \ell(Y_i, 0)$$

is L_ℓ -Lipschitz with respect to the Euclidean norm and vanishes at the origin. The vector-contraction inequality of [Maurer \(2016\)](#) therefore gives

$$\hat{\mathfrak{R}}_{S_0}(\mathcal{G}_{\hat{T}}) \leq \sqrt{2} L_\ell \hat{\mathfrak{R}}_{n_0}^m(\mathcal{F}_{\hat{T}}(B)).$$

The scalar complexity on the left uses the conventional one-sided supremum. The absolute-value vector complexity used in this paper is at least as large as its one-sided counterpart, so the displayed inequality remains valid with our vector-complexity definition on the right.

Define the conditional complexity radius

$$r_{\hat{T}} := \sqrt{2} L_\ell B \|\hat{T}\| \sqrt{\frac{\kappa \operatorname{Tr}(\mathbf{M})}{n_0}}.$$

By [Lemma C.2](#), for every target sample realization,

$$\hat{\mathfrak{R}}_{S_0}(\mathcal{G}_{\hat{T}}) \leq r_{\hat{T}}.$$

Consequently, the conditional expected Rademacher complexity satisfies

$$\begin{aligned} \mathfrak{R}_{n_0}(\mathcal{G}_{\hat{T}} \mid \hat{T}) &:= \mathbb{E}_{S_0 \mid \hat{T}} [\hat{\mathfrak{R}}_{S_0}(\mathcal{G}_{\hat{T}})] \\ &\leq r_{\hat{T}}. \end{aligned}$$

The quantity $r_{\hat{T}}$ is deterministic after conditioning on the source data.

Apply [Lemma C.3](#) conditionally on $\hat{T}$ to the class $\mathcal{G}_{\hat{T}} \subset [0, C_\ell]$. With conditional probability at least $1 - \delta$, the following inequalities hold simultaneously for every $f \in \mathcal{F}_{\hat{T}}(B)$:

$$\begin{aligned} R_0(f) &\leq \hat{R}_0(f) + 2r_{\hat{T}} + C_\ell \sqrt{\frac{\log(2/\delta)}{2n_0}}, \\ \hat{R}_0(f) &\leq R_0(f) + 2r_{\hat{T}} + C_\ell \sqrt{\frac{\log(2/\delta)}{2n_0}}. \end{aligned}$$

Let $\eta > 0$. By the definition of the infimum, there is $f_\eta \in \mathcal{F}_{\hat{T}}(B)$ such that

$$R_0(f_\eta) \leq \inf_{f \in \mathcal{F}_{\hat{T}}(B)} R_0(f) + \eta.$$

Because $\widehat{f}_0$ is an empirical risk minimizer,

$$\widehat{R}_0(\widehat{f}_0) \leq \widehat{R}_0(f_\eta).$$

Apply the first deviation inequality to $\widehat{f}_0$, use empirical optimality, and then apply the second deviation inequality to f_η :

$$\begin{aligned} R_0(\widehat{f}_0) &\leq \widehat{R}_0(\widehat{f}_0) + 2r_{\widehat{T}} + C_\ell \sqrt{\frac{\log(2/\delta)}{2n_0}} \\ &\leq \widehat{R}_0(f_\eta) + 2r_{\widehat{T}} + C_\ell \sqrt{\frac{\log(2/\delta)}{2n_0}} \\ &\leq R_0(f_\eta) + 4r_{\widehat{T}} + 2C_\ell \sqrt{\frac{\log(2/\delta)}{2n_0}} \\ &\leq \inf_{f \in \mathcal{F}_{\widehat{T}}(B)} R_0(f) + \eta + 4r_{\widehat{T}} + 2C_\ell \sqrt{\frac{\log(2/\delta)}{2n_0}}. \end{aligned}$$

Let $\eta \downarrow 0$ and subtract R_0^* . By the definition of $A_{\widehat{T}}(B)$,

$$R_0(\widehat{f}_0) - R_0^* \leq A_{\widehat{T}}(B) + 4r_{\widehat{T}} + 2C_\ell \sqrt{\frac{\log(2/\delta)}{2n_0}}.$$

Insert the definition of $r_{\widehat{T}}$:

$$\begin{aligned} R_0(\widehat{f}_0) - R_0^* &\leq A_{\widehat{T}}(B) + 4\sqrt{2} L_\ell B \|\widehat{T}\| \sqrt{\frac{\kappa \operatorname{Tr}(\mathbf{M})}{n_0}} \\ &\quad + 2C_\ell \sqrt{\frac{\log(2/\delta)}{2n_0}}. \end{aligned}$$

This is the asserted conditional bound. Replacing $\|\widehat{T}\|$ by $\|\widehat{T}\|_{\text{HS}}$ is valid because every Hilbert-Schmidt operator satisfies $\|\widehat{T}\| \leq \|\widehat{T}\|_{\text{HS}}$. $\square$

D A Sufficient Sobolev Activation Condition

The main Sobolev theorems assume boundedness of each activation Koopman operator. The following lemma records a standard sufficient condition in the integer-order case and verifies that the same scalar composition estimate remains valid for the $\mathbf{M}^{-1}$ -weighted vector norm.

Lemma D.1 (Bounded Sobolev activation composition). *Let $s \in \mathbb{N}_{>0}$, let $d, m \in \mathbb{N}_{>0}$, and let $\mathbf{M} \succ 0$. Suppose that $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a C^s bi-Lipschitz diffeomorphism, that every partial derivative $D^\alpha \sigma_j$ with $1 \leq |\alpha| \leq s$ is bounded, and that*

$$\left\| \det D\sigma^{-1} \right\|_{L^\infty(\mathbb{R}^d)} < \infty.$$

Then

$$\mathcal{K}_\sigma : H_{\mathbf{M}}^s(\mathbb{R}^d, \mathbb{R}^m) \longrightarrow H_{\mathbf{M}}^s(\mathbb{R}^d, \mathbb{R}^m)$$

is bounded.

Proof. We first establish the scalar estimate on the dense core $C_c^\infty(\mathbb{R}^d)$. For integer s , the Fourier Sobolev norm is equivalent to the weak-derivative norm. Thus there are constants $c_1, c_2 > 0$, depending only on d, s , and the Fourier normalization, such that every scalar $u \in H^s(\mathbb{R}^d)$ satisfies

$$\begin{aligned} c_1 \sum_{|\alpha| \leq s} \|D^\alpha u\|_{L^2(\mathbb{R}^d)}^2 &\leq \|u\|_{H^s(\mathbb{R}^d)}^2 \\ &\leq c_2 \sum_{|\alpha| \leq s} \|D^\alpha u\|_{L^2(\mathbb{R}^d)}^2. \end{aligned}$$

Fix $u \in C_c^\infty(\mathbb{R}^d)$. We bound every derivative of $u \circ \sigma$ of order at most s . For the zeroth-order term, the change of variables $y = \sigma(x)$ gives

$$\begin{aligned} \|u \circ \sigma\|_{L^2(\mathbb{R}^d)}^2 &= \int_{\mathbb{R}^d} |u(\sigma(x))|^2 dx \\ &= \int_{\mathbb{R}^d} |u(y)|^2 |\det D\sigma^{-1}(y)| dy \\ &\leq \|\det D\sigma^{-1}\|_{L^\infty} \|u\|_{L^2(\mathbb{R}^d)}^2. \end{aligned}$$

Fix a multi-index α with $1 \leq |\alpha| \leq s$. Because both u and σ are classically differentiable to the required orders, the multivariate chain rule expresses $D^\alpha(u \circ \sigma)$ as a finite sum of terms of the form

$$(D^\beta u)(\sigma(x)) P_{\alpha,\beta} \left((D^\gamma \sigma_j(x))_{1 \leq j \leq d, 1 \leq |\gamma| \leq |\alpha|} \right),$$

where $1 \leq |\beta| \leq |\alpha|$ and $P_{\alpha,\beta}$ is a polynomial determined only by the multi-indices. The number of terms is finite and depends only on d and s . By the assumed bounds on the derivatives of σ , every polynomial factor is uniformly bounded. Hence there is $C_\alpha > 0$ such that

$$|D^\alpha(u \circ \sigma)(x)|^2 \leq C_\alpha \sum_{1 \leq |\beta| \leq |\alpha|} \left| (D^\beta u)(\sigma(x)) \right|^2.$$

Integrate over x and apply the same change of variables to every term:

$$\begin{aligned} \|D^\alpha(u \circ \sigma)\|_{L^2}^2 &\leq C_\alpha \sum_{1 \leq |\beta| \leq |\alpha|} \int_{\mathbb{R}^d} \left| (D^\beta u)(\sigma(x)) \right|^2 dx \\ &= C_\alpha \sum_{1 \leq |\beta| \leq |\alpha|} \int_{\mathbb{R}^d} |D^\beta u(y)|^2 |\det D\sigma^{-1}(y)| dy \\ &\leq C_\alpha \|\det D\sigma^{-1}\|_{L^\infty} \sum_{1 \leq |\beta| \leq |\alpha|} \|D^\beta u\|_{L^2}^2. \end{aligned}$$

Sum over all α with $|\alpha| \leq s$. Since only finitely many multi-indices occur, there is $C_\sigma > 0$ such that

$$\sum_{|\alpha| \leq s} \|D^\alpha(u \circ \sigma)\|_{L^2}^2 \leq C_\sigma \sum_{|\beta| \leq s} \|D^\beta u\|_{L^2}^2.$$

Use the norm equivalence in both directions:

$$\|u \circ \sigma\|_{H^s(\mathbb{R}^d)}^2 \leq c_2 \sum_{|\alpha| \leq s} \|D^\alpha(u \circ \sigma)\|_{L^2}^2$$

$$\begin{aligned}
&\leq c_2 C_\sigma \sum_{|\beta| \leq s} \|D^\beta u\|_{L^2}^2 \\
&\leq \frac{c_2 C_\sigma}{c_1} \|u\|_{H^s(\mathbb{R}^d)}^2.
\end{aligned}$$

Define

$$C_{\text{comp}} := \left(\frac{c_2 C_\sigma}{c_1} \right)^{1/2}.$$

We have proved

$$\|u \circ \sigma\|_{H^s} \leq C_{\text{comp}} \|u\|_{H^s}$$

for every $u \in C_c^\infty(\mathbb{R}^d)$.

We now extend the estimate to an arbitrary $u \in H^s(\mathbb{R}^d)$. Choose $u_r \in C_c^\infty(\mathbb{R}^d)$ such that

$$u_r \longrightarrow u$$

in $H^s(\mathbb{R}^d)$. For every $r, q \in \mathbb{N}_{>0}$, the estimate on the smooth core applied to $u_r - u_q$ gives

$$\begin{aligned}
\|u_r \circ \sigma - u_q \circ \sigma\|_{H^s} &= \|(u_r - u_q) \circ \sigma\|_{H^s} \\
&\leq C_{\text{comp}} \|u_r - u_q\|_{H^s}.
\end{aligned}$$

Thus $(u_r \circ \sigma)_{r \in \mathbb{N}_{>0}}$ is Cauchy in $H^s(\mathbb{R}^d)$. Completeness gives a function $v \in H^s(\mathbb{R}^d)$ such that

$$u_r \circ \sigma \longrightarrow v$$

in $H^s(\mathbb{R}^d)$ and therefore in $L^2(\mathbb{R}^d)$.

The same change of variables used for the zeroth-order term gives

$$\begin{aligned}
\|(u_r - u) \circ \sigma\|_{L^2(\mathbb{R}^d)}^2 &= \int_{\mathbb{R}^d} |u_r(y) - u(y)|^2 |\det D\sigma^{-1}(y)| \, dy \\
&\leq \left\| \det D\sigma^{-1} \right\|_{L^\infty} \|u_r - u\|_{L^2(\mathbb{R}^d)}^2 \\
&\longrightarrow 0.
\end{aligned}$$

Hence $u_r \circ \sigma$ converges to $u \circ \sigma$ in L^2 . The L^2 limit is unique, so $v = u \circ \sigma$ almost everywhere. Therefore $u \circ \sigma \in H^s(\mathbb{R}^d)$. Passing to the limit in the smooth-core estimate gives

$$\begin{aligned}
\|u \circ \sigma\|_{H^s} &= \lim_{r \rightarrow \infty} \|u_r \circ \sigma\|_{H^s} \\
&\leq C_{\text{comp}} \lim_{r \rightarrow \infty} \|u_r\|_{H^s} \\
&= C_{\text{comp}} \|u\|_{H^s}.
\end{aligned}$$

Thus scalar composition is bounded on all of $H^s(\mathbb{R}^d)$.

We now pass to the weighted vector-valued space. Let

$$\mathbf{A} := \mathbf{M}^{-1/2}$$

and write $v = \mathbf{A}f$. Since $\mathbf{A}$ is a constant output transformation,

$$\mathbf{A}(f \circ \sigma) = (\mathbf{A}f) \circ \sigma = v \circ \sigma.$$

The weighted norm is

$$\|f\|_{H_{\mathbf{M}}^s(\mathbb{R}^d, \mathbb{R}^m)}^2 = \sum_{a=1}^m \|v_a\|_{H^s(\mathbb{R}^d)}^2.$$

Apply the scalar estimate to every component v_a :

$$\begin{aligned} \|f \circ \sigma\|_{H_{\mathbf{M}}^s}^2 &= \sum_{a=1}^m \|v_a \circ \sigma\|_{H^s}^2 \\ &\leq C_{\text{comp}}^2 \sum_{a=1}^m \|v_a\|_{H^s}^2 \\ &= C_{\text{comp}}^2 \|f\|_{H_{\mathbf{M}}^s}^2. \end{aligned}$$

Taking square roots proves boundedness of $\mathcal{K}_\sigma$ on the weighted vector-valued Sobolev space. $\square$

E Additional Experimental Details

E.1 Proxy regularization on MNIST

The regularization experiment uses the stabilized numerical proxies described in [Section 7](#). These quantities are optimization penalties motivated by the layer geometry of the theory; they are not evaluations of [Theorems 1 to 3](#) for the rank-deficient experimental architecture.

The fully connected network has the form

$$f_\theta(x) = g(W_4 \sigma(W_3 \sigma(W_2 \sigma(W_1 x + b_1) + b_2) + b_3) + b_4),$$

with weight dimensions

$$\begin{aligned} W_1 &\in \mathbb{R}^{1024 \times 784}, & W_2 &\in \mathbb{R}^{2048 \times 1024}, \\ W_3 &\in \mathbb{R}^{2048 \times 2048}, & W_4 &\in \mathbb{R}^{10 \times 2048}. \end{aligned}$$

The bias dimensions are $b_1 \in \mathbb{R}^{1024}$, $b_2 \in \mathbb{R}^{2048}$, $b_3 \in \mathbb{R}^{2048}$, and $b_4 \in \mathbb{R}^{10}$. Thus the hidden widths are 1024, 2048, and 2048, followed by a ten-dimensional output layer.

The matrices W_1 and W_2 use orthogonal initialization ([Saxe et al., 2014](#)); W_3 and W_4 use a truncated-normal initialization. All biases use a uniform initialization. The activation is the smooth Leaky ReLU of [Biswas et al. \(2022\)](#). Training uses 1000 randomly selected MNIST training examples, runs for 1800 epochs, and uses Adam ([Kingma and Ba, 2015](#)) with learning rate 10^{-4} . No additional L_2 weight decay is used.

Let $\mathcal{I} = \{1, 2\}$ denote the layers on which the proxy penalties are evaluated. The Sobolev-inspired and Brownian-inspired penalties are, respectively,

$$\begin{aligned} \text{SR} &:= \sum_{l \in \mathcal{I}} \frac{\|W_l\|^{s_l}}{\det(\mathbf{I} + W_l^\top W_l)^{1/4}}, & s_l &:= \frac{d_l + 0.1}{2}, \\ \text{BR} &:= \sum_{l \in \mathcal{I}} \frac{\|W_l\|}{\det(\mathbf{I} + W_l^\top W_l)^{1/4}}. \end{aligned}$$

Each penalty is added to the empirical training loss with coefficient $\gamma = 0.01$. The identity inside the determinant prevents singularity for rectangular or rank-deficient weights. It also makes these penalties different from the determinant terms in the proved injective bound.

Figure 1(b) reports test accuracy across five independent initializations. In this experiment, the Brownian-inspired penalty produces a higher mean test-accuracy trajectory than the unregularized baseline, whereas the Sobolev-inspired penalty produces a lower trajectory. This is an empirical observation for the stated architecture and hyperparameters, not a general ordering theorem for the two RKHS regimes.